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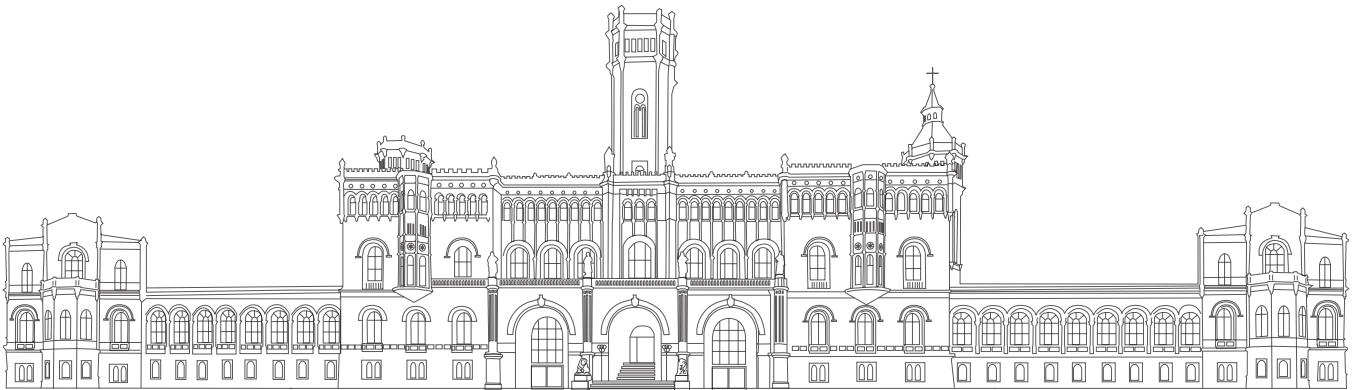


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Sparse Autoencoders for Vision Language Models

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Abstract

As Vision-Language Models (VLMs) become integral to artificial intelligence applications, mitigating the social biases encoded within their foundational architectures is increasingly critical. Traditional debiasing methods rely on computationally expensive fine-tuning or brittle data augmentation. In this thesis, I propose a parameter-efficient, inference-time approach to neutralize occupational gender bias using targeted concept steering via Sparse Autoencoders (SAEs). I evaluate standard Top-K and Batch Top-K SAEs across multiple depths of the LLaVA-1.5 vision encoder to disentangle complex, polysemantic visual representations. To identify specific demographic concepts, I formulate an evaluation pipeline utilizing Signed Mutual Information paired with an Activation Consistency filter. This approach successfully isolates gender-encoding latents in uniform datasets, bypassing the mathematical bottlenecks of continuous semantic metrics.

Through causal intervention experiments, I demonstrate that clamping these isolated features during the forward pass effectively neutralizes severe baseline occupational biases. My findings reveal that the Batch Top-K architecture is crucial for maintaining representational fidelity and that demographic concepts crystallize exclusively in deeper network layers. Furthermore, I show that successful debiasing requires bipolar steering, forcefully pushing the visual representation in the opposite direction to overcome the deep-seated language priors of the Large Language Model backend. Ultimately, this thesis establishes a transparent, highly precise framework for auditing and aligning multimodal architectures with equitable deployment standards without requiring resource-intensive model retraining.

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Chapter 1

Introduction

As artificial intelligence systems become increasingly integrated into society, researchers must understand and mitigate the social biases encoded within foundational architectures. In this thesis, I explore the intersection of VLMs and mechanistic interpretability. I propose a targeted, intervention-based approach to neutralize occupational gender bias without the need for resource-intensive model retraining.

This introductory chapter establishes the foundation for my research. Section 1.1 outlines the primary motivation for this work, detailing the limitations of current debiasing techniques and the promise of SAEs. Section 1.2 formally defines my core research goals and the specific questions this thesis answers. Section 1.3 lists my primary methodological contributions, Section 1.4 provides a high-level overview of my empirical results, and Section 1.5 details the structural outline of the remaining chapters.

1.1 Motivation

VLMs seamlessly blend visual understanding with advanced text generation [22]. However, because developers train these models on massive, uncensored web datasets, the models inherently internalize and propagate severe social biases [31, 30]. For example, when presented with an image of a female professional, VLMs frequently skew their text generation toward stereotypical occupations rather than accurately reflecting the explicit visual context [14]. Traditional methods for mitigating these biases rely on computationally expensive fine-tuning [32], adversarial retraining [16], or brittle data augmentation techniques [3].

Mechanistic interpretability offers a parameter-efficient alternative. I utilize SAEs to disentangle complex, polysemantic activations into inter-

pretable, monosemantic concepts [5, 8]. By algorithmically isolating specific geometric directions representing demographic concepts within the vision encoder, I theoretically neutralize downstream occupational bias through precise, inference-time causal interventions [25]. This approach fundamentally shifts bias mitigation away from opaque retraining pipelines toward transparent, surgical adjustments in the latent space.

1.2 Research Goal

The primary objective of this thesis is to investigate whether SAEs can effectively disentangle demographic representations within the latent space of a VLM, and whether I can causally steer these isolated concepts to mitigate occupational bias. To systematically evaluate this objective, I address the following three research questions:

1. How do different SAE architectures (Top-K vs. Batch Top-K) and varying network depths affect the extraction of demographic concepts in VLMs?
2. How does Signed Mutual Information (MI) compare with the MonoSemanticity (MS) Score when identifying demographic features on highly uniform datasets?
3. Can targeted activation clamping of isolated gender features reduce occupational bias in VLMs without degrading baseline text generation capabilities?

1.3 Contributions

This thesis applies the SAE framework [5] to the domain of social bias mitigation, offering the following empirical contributions:

- **Empirical Comparison of SAE Architectures:** I evaluate standard Top-K and Batch Top-K SAEs across multiple depths of the LLaVA-1.5 vision encoder. I provide an empirical comparison demonstrating that the dynamic capacity allocation of Batch Top-K architectures is significantly more effective at disentangling complex demographic concepts than rigid sparsity constraints.
- **Application of Information-Theoretic Identification:** I utilize an evaluation pipeline combining Signed Mutual Information (MI) and

Activation Consistency (AC) thresholds to isolate demographic concepts. I demonstrate that this discrete statistical approach successfully bypasses the mathematical bottlenecks that cause continuous semantic metrics (like the MS Score) to fail on highly uniform demographic datasets.

- **Mechanisms of Occupational Debiasing:** Building upon the causal steering framework established by [25], I execute inference-time interventions specifically targeted at isolated gender features. I demonstrate that mitigating ingrained occupational bias requires multi-feature, bipolar steering to actively counteract the VLM’s deep-seated language priors, ultimately achieving demographic parity without model retraining.

1.4 Overview of Results

My empirical evaluation reveals that Batch Top-K SAEs outperform standard Top-K architectures when processing complex visual tokens. I successfully localized highly abstracted, steerable demographic concepts in deep network layers (Layer 17 and Layer 22), while shallow layers (Layer 11) failed to yield viable candidates. Furthermore, my causal interventions prove that applying task-specific, bipolar steering to these isolated features neutralizes severe baseline biases. By strictly calibrating the intervention scalar, I achieved near-perfect demographic parity in occupational prediction tasks without inducing logit collapse.

1.5 Outline

The remainder of this thesis unfolds as follows:

- **Chapter 2 (Background and Related Work)** provides the theoretical foundations of VLMs, social bias, mechanistic interpretability, and the operational mechanics of SAEs.
- **Chapter 3 (Methodology)** details my experimental design, including dataset selection, the SAE training pipeline, the formulation of information-theoretic metrics, and the method for targeted causal steering.
- **Chapter 4 (Results)** presents my empirical findings, covering structural SAE validation, demographic concept localization, baseline bias quantification, and overall intervention efficacy.

- **Chapter 5 (Discussion)** interprets these results to directly answer my research questions. I discuss the impact of network depth, metric efficacy, the broader implications for algorithmic fairness, and the limitations of my research alongside potential future research directions.
- **Chapter 6 (Conclusion)** highlights my core findings and summarizes the impact of inference-time concept steering on multimodal architectures.

Chapter 2

Background and Related Work

In this chapter, I first provide the theoretical and methodological foundations necessary for mitigating social bias in Vision-Language Models (VLMs) through targeted concept steering before delving into specific techniques used in this thesis. The chapter is structured as follows:

- In Section 2.1, I introduce modern VLM architectures and examine the mechanics of their multimodal integration.
- Section 2.2 describes the manifestation of social biases in VLMs, highlighting how these models overfit to spurious environmental confounders rather than relying on explicit visual evidence.
- Section 2.3 proceeds to elaborate on Mechanistic Interpretability, a framework used to reverse-engineer neural networks into understandable components.
- In Section 2.4, I then focus specifically on Sparse Autoencoders (SAEs), the primary disentanglement algorithm for Mechanistic Interpretability.
- Finally, Section 2.5 presents the application of SAEs within VLMs for extracting concepts and steering generation. This section also highlights important properties and metrics regarding the evaluation of such algorithms.

2.1 Vision Language Models (VLMs)

Modern Vision-Language Models (VLMs) integrate textual and visual data to perform complex cross-modal reasoning [10]. Earlier VLMs relied on dual-stream architectures like CLIP, which processed visual and language data in

isolated silos and compared them only at the final output stage. While this allowed for strong basic recognition, the lack of interaction between the modalities during the actual processing limited the model’s ability to perform complex, reasoning-heavy tasks. To address this, Large Vision-Language Models (LVLMs) now treat language as a universal interface, guiding an end-to-end trained assistant [22].

Structurally, LVLMs consist of three core components: a pre-trained vision encoder (e.g., CLIP-ViT), a pre-trained LLM (e.g., Vicuna or LLaMA), and a modality interface (often a linear projection layer or Q-Former) that aligns the modalities. The LLaVA architecture serves as a baseline for this multimodal integration [22].

As shown in Figure 2.1, LLaVA connects a pre-trained CLIP visual encoder to a Vicuna LLM decoder via a linear projection matrix. Visual features are mapped into the LLM’s word embedding space, allowing the LLM to process them autoregressively alongside text. While this lightweight integration preserves pre-trained knowledge, forcing visual signals through the LLM bottleneck renders the model susceptible to the LLM’s inherent language priors and statistical skews [17].

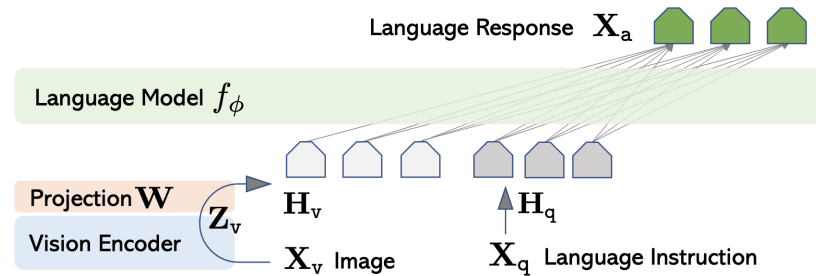


Figure 2.1: The LLaVA network architecture. The model processes images using a CLIP ViT-L/14 encoder, maps the grid features into language embedding tokens via a simple linear projection matrix W , and feeds the combined visual and text tokens into the LLM for autoregressive prediction [22].

2.2 Social Bias in VLMs

VLMs inherit and amplify biases embedded within uncured pre-training datasets [24]. This statistical skew causes models to overfit to spurious environmental confounders. As a result, the models often ignore explicit visual evidence in favor of socially ingrained language priors [21].

Because models optimize for overall statistical efficiency, they latch onto stereotypic co-occurrences. Notable examples from the literature include models consistently misgendering women engaged in sports [16] or misidentifying a purse held by a man as a briefcase [34]. Unsupervised pre-training inherently clusters female-associated tokens with domestic concepts and male tokens with professional concepts [19, 6].

This geometric entanglement is highly detrimental because it propagates severe representational and allocational harms during downstream applications [24]. Representational harms occur when a model demeans, invisibilizes, or stereotypes specific social groups. Allocational harms occur when these biased outputs are used in automated decision-making to unfairly distribute resources or opportunities. For example, when probed with gender-neutral queries, VLMs retrieve images that aggressively diverge from equal demographic distributions. They default to masculine subjects for cognitive or demanding roles and feminine subjects for domestic or service roles [14].

Consequently, deploying these biased models in real-world applications creates a dangerous feedback loop. Rather than reflecting reality, the models act as gatekeepers that reinforce historical inequities, validate harmful cultural stereotypes, and actively alienate marginalized groups from professional and public visibility.

2.3 Mechanistic Interpretability

Mechanistic interpretability aims to reverse-engineer neural networks into understandable algorithms. To achieve this, researchers attempt to decompose the network into smaller, functional units called “features” [8]. A feature represents a distinct semantic concept, and isolating these features allows us to understand the model’s overarching behavior.

Ideally, the most basic structural unit of the network—the individual neuron—would represent a single, understandable feature. However, this is rarely the case. A fundamental roadblock in mechanistic interpretability is polysemanticity [5]. Polysemanticity occurs when a single neuron activates for multiple, completely unrelated concepts. For example, empirical analysis shows that a single vision neuron might fire for both cat faces and the fronts of cars. In language models, one neuron might activate for an incomprehensible mixture of academic citations, English dialogue, and computer code [5]. This polysemantic behavior prevents researchers from using individual neurons to find human-understandable explanations for a network’s decisions [8].

To explain why polysemanticity happens, the literature relies on the Superposition Hypothesis [12]. This theoretical framework states that neural

networks naturally learn significantly more independent features from their training data than they have available neurons. Because a network only has a limited number of dimensions (neurons), it cannot assign a dedicated, independent neuron to every single feature it learns [8].

To solve this space limitation, the network mathematically packs multiple features into an “overcomplete” space. It overlaps these features across multiple neurons in almost-orthogonal directions [12]. The network successfully simulates a much larger model, but when we look at any single neuron, this overlapping manifests as polysemanticity.

Early attempts to resolve superposition focused on changing the network architecture itself. Researchers introduced techniques like Softmax Linear Units (SoLU) or added L_1 regularization during training to force neurons to represent single concepts [11]. While these methods increased the number of interpretable neurons slightly, they consistently failed to eliminate polysemanticity entirely and often degraded the model’s overall performance. Because models inherently benefit from packing more features into limited dimensions, architectural constraints alone are not enough [5].

Consequently, the field has pivoted away from changing how models are trained. Instead, researchers now rely on post-hoc decomposition algorithms. These algorithms are applied after a model is fully trained and are specifically designed to unweave these superimposed representations [28]. This necessity led to the development of Sparse Dictionary Learning methodologies, which are detailed in the following section.

2.4 Sparse Autoencoders (SAEs)

The primary tool used to perform sparse dictionary learning is the Sparse Autoencoder (SAE) [8]. Originally popularized for interpreting Large Language Models (LLMs), an SAE is a secondary, unsupervised neural network trained to act as a microscope on the internal activations of a target model. By projecting the dense, superimposed activations into a higher-dimensional, “overcomplete” latent space, the SAE provides enough room for features to be represented orthogonally as single, independent concepts [5].

An SAE consists of an encoder that maps a dense model activation $\mathbf{x} \in \mathbb{R}^{d_{\text{hidden}}}$ to a sparse latent representation $\mathbf{f} \in \mathbb{R}^{d_{\text{SAE}}}$, where $d_{\text{SAE}} \gg d_{\text{hidden}}$, and a decoder that reconstructs the original activation as $\hat{\mathbf{x}}$ [8]. The baseline architecture utilizes a ReLU activation function to enforce non-negativity and sparsity:

$$\mathbf{f}(\mathbf{x}) = \text{ReLU}(W_{\text{enc}}\mathbf{x} + \mathbf{b}_{\text{enc}}) \quad (2.1)$$

$$\hat{\mathbf{x}}(\mathbf{f}) = W_{\text{dec}}\mathbf{f} + \mathbf{b}_{\text{dec}} \quad (2.2)$$

where W_{enc} and W_{dec} represent the encoder and decoder weight matrices. The columns of the decoder matrix effectively serve as a dictionary of learned feature directions [5].

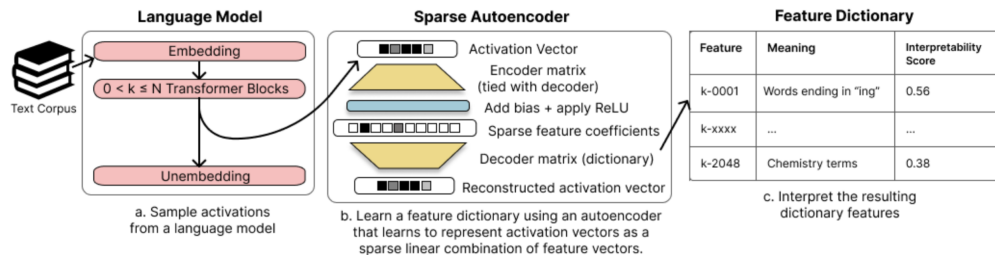


Figure 2.2: An overview of the Sparse Autoencoder methodology. The autoencoder extracts internal network activations, projects them through a sparsity bottleneck to learn an overcomplete feature dictionary, and enables the interpretation of individual monosemantic latents [8].

The standard objective function for an SAE balances two competing goals. It minimizes the reconstruction error between the original activation and the reconstructed activation, ensuring no important information is lost. Simultaneously, it enforces a sparsity penalty to ensure the model utilizes the absolute minimum number of active features necessary for that reconstruction [8]. This is formally defined as:

$$\mathcal{L}(x) = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 + \alpha \|\mathbf{f}\|_1 \quad (2.3)$$

where α is a hyperparameter controlling the L_1 sparsity penalty. By optimizing this loss, the SAE successfully disentangles overlapping signals into a set of sparse, monosemantic features, providing a clear vocabulary for mechanistic analysis [5].

2.4.1 Feature Identification and Interpretation

Once an SAE generates a dictionary of learned features from an LLM, the next challenge is determining the semantic meaning of these discrete latents. The literature outlines several methodological approaches to systematically interpret these concepts.

The most direct method is an input-based approach commonly referred to as *MaxAct*. Because SAE features are intrinsically sparse, a properly disentangled feature vector should only activate for a highly specific conceptual pattern. By observing the text contexts that maximally activate a specific feature, researchers can manually deduce its underlying semantic concept [5].

To verify that a feature functionally represents a concept rather than merely correlating with input text, researchers also rely on output-based explanations. By examining the *logit weights*, one can observe which specific output tokens a feature actively boosts or suppresses. If a feature is truly monosemantic, its causal role in text generation cleanly aligns with its input-based interpretation [8].

However, scaling feature identification across thousands of latents via manual inspection is unfeasible. To address this, automated interpretability pipelines are heavily utilized. This process involves using a separate LLM to generate natural language explanations based on a feature’s highly activating contexts [8]. This yields a quantitative interpretability score, allowing researchers to systematically filter and select high-quality concept features from the overcomplete dictionary.

2.4.2 Concept Steering

Once an overcomplete dictionary of monosemantic features is identified, researchers can use these features to causally verify their functional importance and exert direct control over the LLM’s output [4]. This process is broadly referred to as *concept steering* or activation engineering. It allows for targeted behavioral modifications during inference without requiring computationally expensive fine-tuning of the underlying weights [28].

The primary mechanism for steering via SAEs relies on activation patching. In this method, the model’s internal residual stream is intercepted during the forward pass. The activation values of a specific subset of identified SAE features (F) are artificially altered. A feature might be ablated (clamped to zero) or amplified to a specific magnitude, while the rest of the network remains untouched [8].

Mathematically, the intervention replaces the original residual stream vector $\mathbf{x}_i$ at a given token position i with a patched vector $\mathbf{x}'_i$. This is formulated as:

$$\mathbf{x}'_i = \mathbf{x}_i + \sum_{j \in F} (\bar{\mathbf{c}}_{i,j} - \mathbf{c}_{i,j}) \mathbf{f}_j \quad (2.4)$$

where $\mathbf{c}_{i,j}$ represents the original encoded activation coefficient for feature j , $\bar{\mathbf{c}}_{i,j}$ represents the artificially modified coefficient, and $\mathbf{f}_j$ denotes the corresponding dictionary feature direction [8].

By applying these precise interventions, researchers can causally link specific features to downstream model generation. For instance, empirical research demonstrates that artificially amplifying a monosemantic feature corresponding to “base64” directly forces a language model to generate base64-encoded text, effectively overriding the standard next-token prediction logic

[5]. Similarly, ablating functionally important features predictably suppresses specific token probabilities, providing concrete proof of the feature’s causal role in the network’s reasoning [4].

2.5 SAEs in VLMs

While Sparse Autoencoders (SAEs) were originally developed to disentangle polysemantic neurons in text-based language models, recent literature has successfully adapted these architectures to interpret the continuous, high-dimensional latent spaces of Vision-Language Models (VLMs) [25].

In multimodal architectures such as LLaVA, the visual encoder (e.g., CLIP) projects dense visual signals into a shared embedding space. Because visual patches lack inherent semantic boundaries, these models inherently encode multiple overlapping objects, textures, and contexts within individual neurons [22, 25]. By training an SAE on the activation outputs of a frozen vision encoder, researchers can systematically decompose these dense visual embeddings into a sparse, overcomplete dictionary.

Within this learned dictionary, individual latent directions correspond to discrete, highly monosemantic visual concepts (Figure 2.3). Unlike standard VLM neurons that exhibit severe polysemanticity by firing indiscriminately for overlapping concepts, SAE latents isolate distinct features ranging from low-level color patterns to high-level semantic objects [25].

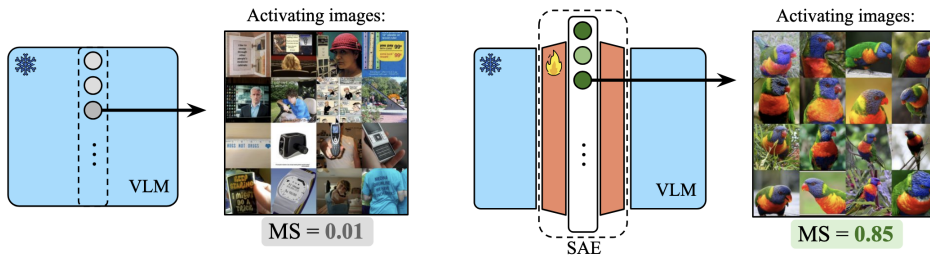


Figure 2.3: Training Sparse Autoencoders over VLMs (e.g., CLIP). The highly activating images of a standard neuron in a pretrained VLM are highly polysemantic (left), whereas a neuron in an SAE trained to reconstruct the same layer is strictly monosemantic, focusing on a single, isolated concept (right) [25].

2.5.1 Feature Identification and Concept Steering

Evaluating feature disentanglement in vision architectures presents unique theoretical differences compared to text-based models. In natural language, individual words frequently require surrounding sentence context to be fully understood. In the visual domain, however, individual images can directly and independently activate specific neurons without additional sequential context [25].

Consequently, feature identification in VLMs heavily relies on analyzing highly activating images. These are the specific images from a dataset that most strongly fire a particular latent direction. Greater semantic similarity among these highly activating images suggests that the neuron is more narrowly focused on a single concept. This consistency allows researchers to accurately assign a semantic label to the isolated SAE neuron.

While the underlying mathematical mechanism of test-time clamping remains identical to text-based models (as detailed in Section 2.4.2), concept steering in VLMs differs fundamentally in its architectural point of intervention. Rather than altering text tokens within the LLM during generation, the intervention targets the visual representations before they bridge into the language module. In architectures like LLaVA, an SAE can be attached to the residual stream of the vision encoder at various targeted depths, ranging from intermediate layers to the final output layer [25].

By intercepting the visual patch tokens at a specific layer, projecting them through the SAE, and clamping a monosemantic latent, researchers alter the visual representation early in the network’s forward pass. When these steered visual tokens complete their processing and are fed into the frozen LLM, the model treats the artificially injected concepts as genuine visual evidence, as depicted in Figure 2.4. Consequently, intervening purely on the residual stream within the vision modality causally dictates and overrides the text modality’s downstream generated response [18, 25].

2.5.2 Structural Evaluation

Evaluating SAEs requires a divided approach to distinguish between training optimization and true semantic utility [29]. Structural metrics evaluate the internal optimization behavior of the SAE during training.

The primary metric for measuring network compression is L_0 sparsity, defined as the average number of active non-zero latents per input [13]. Conversely, the preservation of the original network’s information is measured via reconstruction fidelity. This is predominantly calculated using Mean Squared Error (MSE), which directly compares the original dense activations with

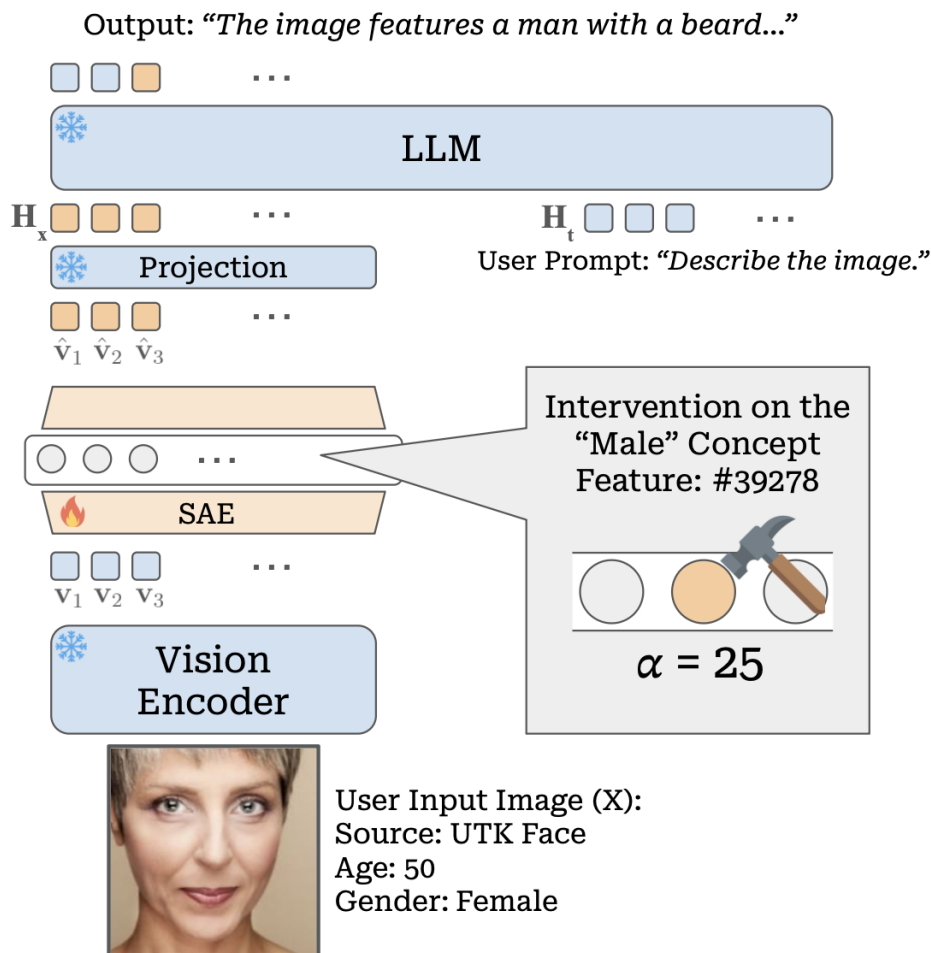


Figure 2.4: Concept steering in a VLM via an SAE. In this example, an SAE is attached to the residual stream of LLaVA’s vision encoder at an intermediate depth (Layer 17). By artificially boosting a monosemantic feature corresponding to the “Male” gender concept, the modified visual tokens successfully override the downstream text generation of the frozen LLM. (Further details and results of these intervention are discussed in Section 4.3).

the SAE-reconstructed activations. Additionally, Explained Variance (R^2) is heavily utilized to measure how much variance in the original data is retained [13]. Optimizing these structural metrics is fundamental to confirming the SAE has trained correctly, but it does not guarantee that the resulting features are functionally useful for bias mitigation.

2.5.3 Functional Evaluation

To verify that the SAE has successfully isolated true semantic concepts and to isolate features for downstream steering, functional metrics are required.

MonoSemanticity (MS) Score The MS metric explicitly measures neuron wise monosemanticity by computing the activation-weighted pairwise similarity of image embeddings [25]. A higher MS indicates that the images yielding the highest activations for a specific neuron share a strictly singular semantic concept.

Mathematically, the formulation relies on a diverse set of images $\mathcal{I} = \{\mathbf{x}_n \in \mathbb{X}\}_{n=1}^N$ and a pretrained deep image encoder E (such as DINOv2 or CLIP) to approximate true semantic similarity. First, the algorithm extracts embeddings to construct a pairwise similarity matrix $\mathbf{S} = [s_{nm}]_{n,m} \in [-1, 1]^{N \times N}$, which captures the cosine similarity between each pair of image embedding vectors:

$$s_{nm} := \frac{E(\mathbf{x}_n) \cdot E(\mathbf{x}_m)}{|E(\mathbf{x}_n)| |E(\mathbf{x}_m)|} \quad (2.5)$$

Next, the activations a_n^k of the k -th SAE neuron across all images are extracted and L_1 -normalized to yield $\tilde{a}_n^k$. These normalized activations are used to compute a relevance matrix $\mathbf{R}^k = [r_{nm}^k]_{n,m} \in [0, 1]^{N \times N}$, which quantifies the shared activation strength for each image pair:

$$r_{nm}^k := \tilde{a}_n^k \tilde{a}_m^k \quad (2.6)$$

Finally, the MonoSemanticity Score for the k -th neuron, $\mathbf{MS}^k \in [-1, 1]$, is derived by calculating the average pairwise semantic similarity, strictly weighted by the relevance matrix, while excluding self-pairings $(\mathbf{x}_n, \mathbf{x}_n)$:

$$\mathbf{MS}^k := \frac{1}{N(N-1)} \sum_{n=1}^N \sum_{\substack{m=1 \\ m \neq n}}^N r_{nm}^k s_{nm} \quad (2.7)$$

By relying on continuous activation magnitudes rather than an arbitrary cutoff of top images, the MS score provides a highly rigorous assessment

of visual monosemanticity that strongly correlates with human conceptual perception [25].

Signed Mutual Information (MI) While the MS Score identifies general conceptual purity, isolating specific demographic features requires precise statistical dependence metrics. Mutual Information (MI) serves as a robust information-theoretic statistic for detecting relationships between continuous data distributions (SAE activation magnitudes) and discrete datasets (binary demographic labels) [27].

Formally, the Mutual Information between a continuous feature activation f_i and a discrete class label y is defined as:

$$\text{MI}(f_i, y) = \sum_{f_i} \sum_y p(f_i, y) \log \frac{p(f_i, y)}{p(f_i)p(y)} \quad (2.8)$$

To utilize this for targeted concept steering, researchers must account for the specific direction of the association. This is achieved by computing the Signed Mutual Information, which multiplies the base MI by the sign of the difference in mean activations between the two target classes:

$$\text{Signed-MI}(f_i) = \text{MI}(f_i, y) \cdot \text{sign}(\Delta \bar{a}_i) \quad (2.9)$$

where $\Delta \bar{a}_i$ represents the difference in the SAE encoded mean activation of feature i between the target groups. In the context of binary gender features, the discrete labels are mathematically adapted into a bipolar format. This adaptation allows the Signed-MI to cleanly segregate the feature space into positive values (e.g., female-associated concepts) and negative values (e.g., male-associated concepts).

Activation Consistency Filter (τ) Relying solely on Signed-MI is insufficient for robust feature selection. A fundamental flaw arises when an SAE latent activates exceptionally strongly for only a trivial number of images. For example, an anomaly where a neuron fires heavily for only two specific women in the entire dataset will yield a high MI score, but it is functionally useless for generalized concept steering.

To resolve this, features must pass an Activation Consistency (AC) filter. This ensures a latent activates reliably across a broad, representative sample of the demographic class. For a feature f_i and a given class c , the activation frequency is computed as:

$$\text{freq}_i^c = \frac{1}{N_c} \sum_{j=1}^{N_c} \mathbb{1}[a_{i,j}] \quad (2.10)$$

where $a_{i,j}$ is the SAE encoded activation of feature i for sample j in class c , N_c is the total number of samples in that class, and $\mathbb{I}[\cdot]$ is the indicator function denoting whether the feature is active.

To isolate class-consistent features, a strict threshold (τ) is applied:

$$\text{Consistent}_i^c = \begin{cases} 1, & \text{if } \text{freq}_i^c \geq \tau \\ 0, & \text{otherwise} \end{cases} \quad (2.11)$$

By defining a threshold, researchers algorithmically eliminate sporadic noise and highly specialized outlier neurons. Furthermore, filtering out features that consistently activate for both classes guarantees that the resulting dictionary elements are highly discriminative and stable for targeted bias intervention.

Chapter 3

Methodology

In this chapter, I describe the methodology chosen to investigate and mitigate social biases within VLMs through targeted concept steering. Building upon the foundational theories of mechanistic interpretability and multimodal architectures established in Chapter 2, this methodology shifts the focus toward practical interventions. After identifying current challenges in VLM debiasing, I establish how SAEs can overcome these limitations by facilitating direct test-time interventions in the latent space. These theoretical approaches are then evaluated empirically through carefully designed experiments.

The chapter is structured as follows:

- Section 3.1 explains the current challenges in VLM debiasing techniques in order to identify avenues for improvement, serving as the foundation for the subsequent sections.
- Section 3.2 explores Concept Steering via SAEs as an intervention strategy to mitigate model bias without the need for resource intensive re-training.
- Section 3.3 provides justifications for the choice of datasets, VLM architecture, SAE variants, and hyperparameters that are applicable to the experiments.
- Section 3.4 describes the detailed steps to carry out the individual experiments, establishing the reproducible pipeline required to identify gender-concept features and evaluate steering efficacy.

3.1 Challenges in VLM debiasing

Developing effective mitigation strategies for VLMs remains a methodological challenge. As established in Section 2.2, biases in these architectures are particularly difficult to isolate because they stem from the individual language and vision encoders, alongside their complex interaction dynamics.

Historically, debiasing efforts have relied on macroscopic interventions categorized into: data-level modifications, in-processing methods, and post-processing adjustments [6, 21]. While successful in smaller unimodal contexts, these approaches present severe practical shortcomings when applied to modern VLMs. They primarily struggle to disentangle the high-dimensional embedding space without compromising overall model utility.

At the data level, applying Counterfactual Data Augmentation (CDA) to the visual domain introduces significant complexities. Modifying a protected attribute directly alters the pixel-level structure. Consequently, naive pre-processing techniques, such as blurring faces or applying blackout boxes (depicted in Figure 3.1), create unnatural image distributions. Models frequently exploit these artifacts as spurious correlations rather than learning true debiasing signals [21].

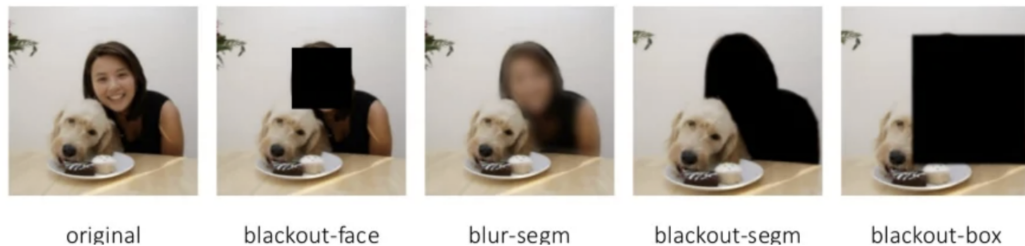


Figure 3.1: Examples of unnatural pre-processing methods attempting to remove sensitive gender attributes from images, which often introduce novel spurious correlations [21].

Similarly, in-processing and retraining pipelines lack scalability. Updating the parametric knowledge of massive foundational models requires immense computational resources and time. Furthermore, global fine-tuning fails to address the root geometric entanglement of concepts. This limitation frequently results in a “Whac-A-Mole” dynamic, where mitigating one specific social bias inadvertently degrades performance on downstream tasks or forces the network to redistribute its reliance onto other overlapping confounders [15].

Therefore, attempting to debias VLMs by manipulating training data, adjusting global loss functions, or applying surface-level prompt instructions

proves insufficient. Overcoming these limitations requires a shift toward mechanistic interpretability. This establishes Gender-Concept Steering via Sparse Autoencoders (SAEs) as a highly precise, parameter-efficient alternative to neutralize targeted biases directly at the inference stage.

3.2 Gender-Concept Steering via SAEs

To overcome the limitations of macroscopic debiasing outlined in Section 3.1, this methodology proposes targeted concept steering via Sparse Autoencoders (SAEs). Instead of attempting to unlearn biases by globally updating model weights or altering training data, this approach shifts the intervention directly to the model’s latent space.

In VLMs, distinct visual and semantic concepts are often forced into highly entangled, overlapping representations within the vision encoder [25]. For the purpose of this research, this architectural bottleneck is highly problematic, as it forces distinct demographic contexts into single, polysemantic neurons. By applying a sparse dictionary learning framework, this methodology disentangles these representations to isolate specific latent directions.

Once these demographic concepts are geometrically isolated within the SAE’s expanded dictionary, they can be directly manipulated during the forward pass. This process, defined as concept steering, intercepts the visual token embeddings and selectively neutralizes the specific activations responsible for the targeted bias. An overview of this architectural flow is depicted in Figure 2.4. The semantically altered tokens are subsequently passed to the frozen Large Language Model (LLM) backend, deterministically steering its generation trajectory away from stereotypical associations.

This surgical manipulation offers a highly efficient mechanism for neutralizing demographic disparities. Recent applications demonstrate that steering specific concepts in this manner removes targeted biases without degrading primary visual fidelity or reasoning capabilities [18]. Furthermore, because the SAE operates as an unsupervised post-hoc attachment, this framework requires zero resource-intensive retraining of the foundational VLM and avoids the introduction of computationally expensive Low-Rank Adapters (LoRAs) [18].

3.3 General Design Choices for Experiments

Prior to delving into the specific experimental procedures, I describe the fundamental design choices that establish the framework for my experiments.

To successfully isolate and mitigate gender bias via concept steering, several interacting components must be carefully defined.

The datasets chosen for training the SAEs and subsequently evaluating the interventions are described in Subsection 3.3.1, while the targeted Vision-Language Model architecture is justified in Subsection 3.3.2. The specific Sparse Autoencoder variants employed for latent disentanglement are explained in Subsection 3.3.3, followed by an outline of their critical hyperparameter configurations in Subsection 3.3.4. Because this methodology relies on precisely identifying gender features and validating their behavioral impact, Subsection 3.3.5 defines the comprehensive suite of evaluation metrics utilized, including the integration of Signed Mutual Information. Finally, Subsection 3.3.6 formalizes the exact mathematical mechanism of steering using the trained SAEs. This establishes the necessary technical foundation for the detailed experimental pipeline presented in Section 3.4.

3.3.1 Datasets

To isolate and mitigate gender bias using Concept Steering via SAEs, I rely on a dual-dataset approach. I require a broad dataset to train the SAEs for general latent disentanglement, alongside a specialized demographic dataset to accurately compute functional evaluation metrics like the Monosemanticity (MS) Score and Signed Mutual Information (MI). Consequently, I utilize the ImageNet dataset [9] and a custom split of the UTKFace dataset [33].

ImageNet for Feature Extraction and Structural Evaluation

I use the ImageNet (ILSVRC 2012) dataset [9], comprising over 1.28 million training images and 50,000 validation images, as the foundational corpus for SAE training and structural evaluation. Its scale and diversity provide a sufficiently rich activation space for me to train the SAE variants, ensuring the autoencoders learn a comprehensive overcomplete dictionary of features.

Specifically, I utilize the ImageNet training split solely to extract residual stream activations from the VLM for SAE training. Following the training phase, I employ the ImageNet validation split to perform baseline structural evaluations: Mean Squared Error (MSE) of the reconstructed activations, variance explained (R^2), and L_0 sparsity. Furthermore, I use the validation split as a proxy to compute the initial Monosemanticity (MS) Scores, which allows me to benchmark the trained dictionaries against values reported in contemporary mechanistic interpretability literature [25].

UTKFace for Concept Identification and Intervention Testing

While ImageNet is optimal for general dictionary learning, I require a dataset with explicit demographic annotations to identify distinct, interpretable gender concepts. I leverage the UTKFace dataset [33] for this purpose, as it is widely used for measuring stereotypical biases in the vision modality [17].

To accurately calculate functional evaluation metrics, I curate a custom split of the UTKFace dataset. I process the formal partitioning of this dataset, defined as $\mathcal{D}_{UTK}$, through the following methodological steps:

1. **Age Filtering:** Following the methodology established by [17], I constrain the dataset to individuals aged 18 to 65. This filters out minors and the elderly, focusing my analysis on the demographic bracket most commonly associated with occupational-social gender biases. This yields a base filtered set $\mathcal{D}_{filter}$ of 4,853 images.
2. **Steering Test Split:** I randomly sample a balanced subset of 1,000 images from $\mathcal{D}_{filter}$ to form the final steering test set, denoted as $\mathcal{D}_{test}$. To prevent distribution skew during the intervention phase, I ensure this set strictly contains 500 images of males and 500 images of females.
3. **Feature Identification Split:** I assign the remaining 3,853 images to constitute the identification set, $\mathcal{D}_{ident}$.

I formalize this custom partition as follows:

$$\mathcal{D}_{filter} = \{x \in \mathcal{D}_{UTK} \mid 18 \leq \text{age}(x) \leq 65\} \implies |\mathcal{D}_{filter}| = 4853 \quad (3.1)$$

$$\mathcal{D}_{filter} = \mathcal{D}_{test} \cup \mathcal{D}_{ident}, \quad \mathcal{D}_{test} \cap \mathcal{D}_{ident} = \emptyset \quad (3.2)$$

I exclusively use the images in $\mathcal{D}_{ident}$ to calculate the MS Score and the Signed Mutual Information (MI) Score. These metrics are critical for me to isolate the specific latent directions that act as gender concepts within the VLM’s residual stream. Once I map these specific features, I utilize $\mathcal{D}_{test}$ to evaluate the efficacy of the concept steering intervention.

3.3.2 VLM Architecture

Two primary constraints govern my selection of the target Vision-Language Model (VLM): the necessity for an open-source architecture that permits unrestrained access to internal activations, and the requirement to establish a comparative baseline against recent mechanistic interpretability literature. Consequently, I select LLaVA 1.5 (Large Language and Vision Assistant) as

the primary architecture for this experimental design. Utilizing LLaVA 1.5 provides the transparency required for deep latent interventions. Furthermore, it aligns directly with the methodology established by [25], allowing me to rigorously compare the efficacy of my concept steering against state-of-the-art findings.

LLaVA connects a pre-trained vision encoder to a Large Language Model (LLM) backend [22]. Specifically, the architecture utilizes the open-set CLIP ViT-L/14 (Vision Transformer) as its visual feature extractor, coupled with the LLaMA-based Vicuna decoder as its language reasoning engine [22].

To bridge the modality gap, LLaVA employs a trainable linear projection matrix, W , which maps the visual grid features extracted by CLIP into the word embedding space of the LLM [22]. For an input image X_v , the visual encoder computes the feature representation $Z_v = g(X_v)$. I derive the language embedding tokens H_v via this projection:

$$H_v = W \cdot Z_v \quad (3.3)$$

The LLM then processes this sequence of visual tokens autoregressively alongside standard textual prompts [22]. Figure 2.1 illustrates the architectural flow of this multimodal alignment.

Patch Tokens vs. CLS Token Processing A critical nuance in LLaVA’s visual processing lies in its handling of the Vision Transformer’s output tokens. Standard ViT architectures process images by splitting them into sequential grid patches alongside an overarching classification (CLS) token [22]. However, to maintain dense, spatial visual context for complex generative tasks, LLaVA relies exclusively on the unpooled spatial patch tokens. Furthermore, the model extracts these grid features strictly before the final processing layer of the vision tower (Layer 23) [22, 25]. By focusing on these intermediate patch tokens, the LLM retains a highly granular, localized representation of the image. While this granularity is vital for instruction-following tasks, it simultaneously exposes the model to the localized representational biases I aim to mitigate.

Layer Selection for Feature Extraction and Intervention Because the LLM effectively utilizes the grid features prior to the final layer, attempting to steer the model using the absolute final outputs or the CLS token would be structurally ineffective. Therefore, to properly isolate and intervene upon gender concepts, I target specific intermediate layers within the vision tower.

Specifically, I extract activations and execute concept steering at Layers 11, 17, and 22 of the CLIP ViT-L/14 encoder. Following the empirical validation by [25], I chose these layers because they capture a spectrum of semantic abstraction. They progress from mid-level texture representations in Layer 11 to highly complex, monosemantic identity concepts in Layer 22.

To implement interventions dynamically at these targeted depths without permanently altering the foundational model weights, I employ PyTorch [2] forward hooks. These hooks allow me to seamlessly intercept the residual stream during the forward pass. This enables the Sparse Autoencoder (SAE) to extract the token activations, clamp the targeted gender feature, and reinject the steered vector into the VLM’s pipeline at runtime.

3.3.3 Sparse Autoencoders (SAEs)

To isolate monosemantic concepts from the polysemantic representations within the VLM’s residual stream, I rely on Sparse Autoencoders (SAEs). Standard SAE architectures have documented theoretical limitations, specifically regarding activation shrinkage and inefficient sparsity allocation [13]. Consequently, I employ both Top-K and Batch Top-K SAE variants to rigidly control sparsity and optimize reconstruction fidelity.

As established in Chapter 2, standard SAEs utilize an L_1 -norm penalty to enforce sparsity. A fundamental limitation of this formulation is “shrinkage bias” [13]. The L_1 penalty systematically shrinks all positive feature activations toward zero, which inherently compromises reconstruction accuracy and underestimates the true magnitude of the underlying concepts.

Top-K Sparse Autoencoders

To directly circumvent the L_1 shrinkage bias, I utilize the Top-K SAE variant. Top-K SAEs replace the L_1 penalty and ReLU thresholding by applying a hard Top-K activation function that isolates only the k highest magnitude latents, aggressively clamping all other activations to zero [13]. I define the encoder operation mathematically as:

$$\mathbf{z} = \text{TopK}(W_{enc}(\mathbf{x} - \mathbf{b}_{pre})) \quad (3.4)$$

where $\mathbf{b}_{pre}$ serves as a pre-encoder bias term. Because the activation function explicitly enforces an exact L_0 sparsity (exactly k active features per token), the problematic L_1 regularization term is entirely eliminated from the loss function [13]. The training objective simplifies to minimizing the reconstruction loss, supplemented by a small auxiliary loss (L_{aux}) to prevent the collapse of inactive or “dead” latents:

$$\mathcal{L} = ||\mathbf{x} - \hat{\mathbf{x}}||_2^2 + \alpha L_{aux} \quad (3.5)$$

Batch Top-K Sparse Autoencoders

While Top-K SAEs effectively resolve feature shrinkage, they introduce a rigid limitation: they force every individual token to activate exactly k features, regardless of the inherent complexity of the input [7]. In the visual domain, an uninformative background patch token should mathematically require fewer active features to reconstruct than a highly complex patch containing overlapping semantic objects.

To address this, I additionally utilize the Batch Top-K SAE variant. Batch Top-K relaxes the token-level k constraint to a batch-level constraint [7]. For a given batch of size n , the algorithm selects the top $n \times k$ activations across the entire batch rather than k activations per individual sample. I formulate the batch-level loss as:

$$\mathcal{L}(\mathbf{X}) = ||\mathbf{X} - \text{BatchTopK}(W_{enc}\mathbf{X} + \mathbf{b}_{enc})W_{dec} + \mathbf{b}_{dec}||_2^2 + \alpha L_{aux} \quad (3.6)$$

Figure 4.1 shows the distribution of latents in one of my trained Batch Top-K SAE. This architectural enhancement allows the model to dynamically allocate feature capacity. By utilizing a high number of latents for information-dense features while assigning minimal latents to simpler inputs, I drastically improve the Mean Squared Error (MSE) without sacrificing the average baseline sparsity [7].

Implementation and Training Scope

I execute the implementations of both the standard Top-K and Batch Top-K SAEs using the open-source `dictionary_learning` library [23]. This repository natively supports the necessary constrained optimization parameters and auxiliary loss integration required to train advanced dictionary learning architectures without substantial engineering overhead.

During the training phase on the Vision-Language Model, passing entire dense images through the network and subsequently computing SAE losses over every distinct spatial patch is computationally prohibitive. Furthermore, it is prone to overfitting on highly correlated adjacent patches. Therefore, I adopt the optimized sampling strategy proposed by [25], which involves extracting and training on only two random visual patch tokens per image during the forward pass. This stochastic sampling forces the SAE to generalize its feature dictionary across the broader distribution of visual concepts without artificially inflating training duration.

3.3.4 Hyperparameters

To accurately identify and steer gender concepts without severely degrading the base model’s general representational fidelity, I must carefully calibrate the structural hyperparameters of the Sparse Autoencoders (SAEs). Achieving an optimal balance between interpretability (monosemanticity) and reconstruction quality (R^2) is highly sensitive to the capacity of the dictionary and the rigidity of the imposed sparsity penalty. Consequently, I derive the hyperparameters for both the Top-K and Batch Top-K SAE variants directly from the empirical visual-modality configurations established by [25].

Expansion Factor (ϵ) To project dense, superimposed activations into a higher-dimensional, overcomplete dictionary, I expand the original residual stream dimension by an expansion factor of $\epsilon = 64$. A sufficiently large expansion factor is critical for disentangling highly complex, overlapping visual signals. Empirical evaluations demonstrate that higher expansion factors directly correlate with an increase in the Monosemanticity (MS) Score of the resulting latent units. This is because a wider dictionary offers the necessary geometric capacity to assign dedicated, orthogonal dimensions to distinct semantic concepts.

Sparsity Constraint (K) In the Top-K architectures, the K parameter explicitly dictates the maximum number of non-zero latent neurons allowed to activate per forward pass. In this methodology, I fix the sparsity parameter to $K = 20$. The choice of K represents a fundamental trade-off between the interpretability of the features and the reconstruction quality of the autoencoder. Imposing an extremely strict sparsity constraint (e.g., $K = 1$) effectively decomposes the representation into highly monosemantic features, but it severely cripples the model’s ability to accurately reconstruct the original input, causing the Explained Variance (R^2) to drop drastically. Conversely, setting K too high introduces polysemanticity by allowing multiple entangled features to fire simultaneously. Fixing $K = 20$ optimally balances these competing objectives, yielding highly interpretable dictionary elements while maintaining a stable reconstruction fidelity.

Optimization Parameters For the optimization phase, I utilize the Adam optimizer to minimize the reconstruction loss. Following the specific architectural guidelines, I train the SAEs over 110,000 optimization steps using minibatches of size 4,096. Furthermore, to prevent parameter oscillation and ensure stable convergence within the highly overcomplete latent space, I scale the learning rate (η) inversely relative to the width of the autoencoder (ω).

Following the original scaling laws proposed by [13], I initialize the learning rate mathematically as:

$$\eta = \frac{16}{125\sqrt{\omega}} \quad (3.7)$$

By adhering to this established hyperparameter regime, I ensure that the dictionaries generated during my training phase possess the requisite sparsity and representational depth to facilitate successful downstream concept steering interventions.

3.3.5 Evaluation Metrics

To assess the performance of the trained Sparse Autoencoders (SAEs) and guarantee the functional validity of the isolated gender concepts, I leverage the bifurcated evaluation framework established in Chapter 2. In this section, I detail exactly how I deploy these structural and functional metrics across the selected dataset partitions to filter and select optimal intervention candidates.

Before conducting semantic analysis, I must verify that the Top-K and Batch Top-K SAE variants successfully learned an efficient, overcomplete representation of the VLM’s residual stream without catastrophic information loss. I compute the structural metrics (Mean Squared Error (MSE), Explained Variance (R^2), and L_0 Sparsity) exclusively on the ImageNet validation split. By extracting activations from the frozen LLaVA vision tower (Layers 11, 17, and 22) across this validation set, I measure the MSE between the dense original activations and the sparse reconstructions. I confirm that the optimization objectives are met by ensuring the R^2 remains stable while the L_0 norm (active features per token) remains strictly constrained.

Following structural validation, I calculate the MonoSemanticity (MS) Score across the ImageNet validation set to evaluate the general conceptual disentanglement achieved by the SAEs. I use the MS Score exclusively as a high-level benchmarking tool. By comparing the MS Score distribution from my trained models against the established baselines reported by [25], I quantitatively verify that the learned dictionaries successfully separate general visual concepts out of superposition.

While effective for evaluating broad visual purity on ImageNet, I identified a critical methodological limitation when applying the MS Score to demographic features. The MS Score relies on a continuous gradient of semantic cosine similarity between image pairs. However, when applied to the heavily curated UTKFace identification split ($\mathcal{D}_{ident}$), this underlying similarity matrix degrades. Because the images in this partition are uniformly

cropped faces differing primarily in gender presentation, the cosine similarity computation mathematically collapses into a near-binary state, evaluating to approximately 1 for same-gender pairs and near 0 for cross-gender pairs. This binarization strips the MS Score of its intended granularity, rendering it an insensitive mechanism for isolating targeted demographic latents. Consequently, I shift to purely information-theoretic metrics to identify the final steering candidates.

Targeted Feature Identification via Signed Mutual Information To isolate the specific latent directions corresponding to male and female visual concepts, I calculate Signed Mutual Information (MI) exclusively on the UTKFace identification split ($\mathcal{D}_{ident}$).

To adapt the base MI formulation established in Section 2.5.3 for targeted directional steering, I map the discrete demographic labels of $\mathcal{D}_{ident}$ into a bipolar format. For every image sample $x^{(j)}$, I transform the label $y^{(j)}$ as follows:

$$y^{(j)} = \begin{cases} -1, & \text{if } x^{(j)} \text{ is labeled Male} \\ 1, & \text{if } x^{(j)} \text{ is labeled Female} \end{cases} \quad \forall x^{(j)} \in \mathcal{D}_{ident} \quad (3.8)$$

I segregate the latent space by multiplying the standard Mutual Information score of a feature by the sign of the difference in its mean activation between these two classes. Therefore, a highly positive Signed-MI value denotes a strong, exclusive association with female-presenting concepts, whereas a highly negative value denotes a strong association with male-presenting concepts.

Applying the Activation Consistency Filter As established in Section 2.5.3, Signed-MI must be paired with an Activation Consistency (AC) filter to eliminate outlier neurons. To execute this filtering on $\mathcal{D}_{ident}$, I evaluate the activation frequency of each feature against a candidate set of strict consistency thresholds, specifically $\tau \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$. Due to the auxiliary loss (L_{aux}) introduced during SAE training, I mark a feature as active only if its encoded activation magnitude strictly exceeds a predefined noise-floor threshold.

Furthermore, to guarantee discriminative power during the intervention phase, I enforce a mutual exclusivity constraint. I add a feature f_i to the final candidate steering set only if it surpasses the consistency threshold τ for one gender class while remaining strictly below τ for the opposing class:

$$\begin{aligned} \text{Candidates} = \{f_i \mid & (\text{freq}_i^{\text{female}} \geq \tau \wedge \text{freq}_i^{\text{male}} < \tau) \\ & \vee (\text{freq}_i^{\text{male}} \geq \tau \wedge \text{freq}_i^{\text{female}} < \tau)\} \end{aligned} \quad (3.9)$$

By applying the Signed-MI calculation and this rigid mutual exclusivity filter across the targeted VLM layers, I systematically eliminate general polysemantic neurons and sporadic noise. This filtering pipeline yields a highly specific, robust set of monosemantic gender-concept features prepared for causal intervention.

3.3.6 Steering Setup

Having isolated and filtered the specific demographic latents using Signed Mutual Information and the Activation Consistency threshold, I deploy these features to causally alter the model’s behavior. To achieve this, I employ a targeted test-time intervention mechanism known as concept steering [25].

During the forward pass, the Vision-Language Model (VLM) transforms the input image into a sequence of continuous visual patch token embeddings. For an individual patch token $\mathbf{v}_i \in \mathbb{R}^d$, the trained Sparse Autoencoder (SAE) projects this dense representation into a high-dimensional, sparse activation vector $\mathbf{a}_i = \phi(\mathbf{v}_i) \in \mathbb{R}^\omega$. I execute the targeted intervention within this sparse, disentangled space.

While some contemporary activation engineering techniques rely on additive steering, where a scaled concept vector is linearly added to the residual stream [18], I utilize a rigid clamping mechanism. Additive steering shifts the representation relative to its baseline, meaning the final activation magnitude still partially depends on the original input image. Conversely, clamping strictly enforces a predefined magnitude, which completely overrides the model’s natural perception of the target concept.

Rather than targeting a singular latent index, I apply the intervention to a subset of candidate gender-concept features, denoted as $K \subset \{1, \dots, \omega\}$. For each targeted latent $j \in K$, I clamp its natural activation to a specific, independently optimized intervention scalar α_j . All other feature activations outside of subset K remain unchanged. I formalize this multi-feature clamping operation mathematically as follows:

$$\hat{a}_i^j = \begin{cases} \alpha_j, & \text{if } j \in K \\ a_i^j, & \text{if } j \notin K \end{cases} \quad \forall j \in \{1, \dots, \omega\} \quad (3.10)$$

The steering parameters α_j serve as my primary dials for controlling the intensity of the intervention. By independently clamping this subset of features, I deterministically inject, amplify, or suppress the isolated demographic attributes within the token’s representation.

Once I manipulate the sparse activation vector, I pass it through the SAE decoder to reconstruct the steered visual token embedding:

$$\hat{\mathbf{v}}_i = \psi(\hat{\mathbf{a}}_i) \in \mathbb{R}^d \quad (3.11)$$

The VLM’s standard projection matrix subsequently processes these dimensionally restored, semantically altered token embeddings $\hat{\mathbf{v}}_i$ and feeds them into the Large Language Model (LLM) backend for autoregressive text generation. As illustrated in Figure 2.4, this mechanism ensures that the LLM conditions its linguistic output on the artificially introduced demographic concept rather than the ground-truth visual data [25]. Crucially, because this pipeline manipulates intermediate activations dynamically via PyTorch forward hooks, I execute the intervention entirely at inference time without requiring any fine-tuning of the foundational LLaVA weights.

3.4 Detailed Experiment Design

With the foundational design choices established, I now detail the procedural execution of my experiments. The objective of this section is to operationalize the theoretical frameworks to empirically investigate and mitigate ingrained demographic biases.

This section describes the step-by-step experimental pipeline required to robustly evaluate concept steering as a targeted debiasing mechanism. I divide the experimental design into two primary, sequential phases:

- **Feature Isolation and Verification (Subsection 3.4.1):** I first describe the procedure for extracting residual stream activations and training the Top-K and Batch Top-K SAE architectures across the selected layers of the vision tower. This subsection details how I apply the structural and information-theoretic filtering pipelines to systematically isolate specific, monosemantic gender concepts from the broader latent dictionary.
- **Causal Intervention and Optimization (Subsection 3.4.2):** Subsequently, I outline the deployment of the steering setup. This encompasses formulating a targeted inference prompt designed to quantify occupational bias, executing the causal intervention via latent activation clamping, and applying a systematic search to the intervention scalar (α). My goal in this phase is to determine the optimal steering configuration for neutralizing the model’s inherent demographic skew.

By strictly adhering to this reproducible pipeline, I ensure that the subsequent causal interventions are both statistically sound and functionally interpretable, effectively bridging the gap between theoretical feature identification and practical bias mitigation.

3.4.1 Feature Isolation and Verification

The first phase of the experimental design extracts the latent representation space of the Vision-Language Model (VLM), maps it to an overcomplete dictionary, and systematically filters that dictionary to isolate robust gender concepts. This process bridges the gap between general geometric disentanglement and targeted demographic identification.

Step 1: Activation Extraction and SAE Training To analyze the hierarchical emergence of features within LLaVA 1.5, I extract residual stream activations from the frozen CLIP ViT-L/14 vision encoder at the previously defined depths: Layers 11, 17, and 22. Using the ImageNet training split, I pass the images through the VLM and intercept the visual patch tokens. I then train both the Top-K and Batch Top-K Sparse Autoencoder variants on the extracted activations from each layer, maintaining the strict hyperparameter configurations established in Section 3.3.4.

This process yields a total of six distinct SAE models for evaluation, defined by the set $\mathcal{M}$:

$$\mathcal{M} = \{11, 17, 22\} \times \{\text{TopK}, \text{BatchTopK}\} \implies |\mathcal{M}| = 6 \quad (3.12)$$

Step 2: Structural Verification and Baseline Disentanglement Before searching for specific demographic features, I must empirically verify that the six trained SAEs successfully achieved general geometric disentanglement.

I execute a forward pass of the activations from ImageNet validation split through the trained models to compute the structural metrics: Mean Squared Error (MSE), Explained Variance (R^2), and L_0 sparsity. I verify that the R^2 remains stable across the six models to confirm the reconstruction objective has been met. Following this, I compute the MonoSemanticity (MS) Score across the entire validation set to benchmark the overall dictionary purity. I prune any models that exhibit catastrophic R^2 collapse or near-zero average MS Scores, deeming them unfit for downstream causal intervention.

Step 3: Feature Isolation via Information-Theoretic Filtering Having verified the structural integrity of the dictionaries, I shift the experimental pipeline to isolate the specific latent directions that encode perceived binary gender concepts. For this step, I exclusively utilize the 3,853 images comprising the custom UTKFace identification split ($\mathcal{D}_{ident}$).

I process $\mathcal{D}_{ident}$ through the valid SAE models and compute the Signed Mutual Information (Signed-MI) for every latent feature. I rank these fea-

tures based on their absolute Signed-MI magnitude to quantify their statistical dependence on the perceived binary gender labels [27].

To prevent the selection of statistical anomalies, I subject the top-ranked Signed-MI features to the Activation Consistency (AC) grid search. I evaluate the activation frequency of each candidate feature across the five progressive threshold intervals defined in Section 3.3.5. I isolate the final intervention candidates by selecting the features that maintain the highest Signed-MI scores while surviving the mutual exclusivity constraint at the strictest possible τ threshold. These verified, highly discriminative monosemantic latents map directly into Phase 2 of the experiment.

3.4.2 Causal Intervention and Optimization

The second phase of the experimental design transitions from passive feature identification to active-causal intervention. Having isolated a robust set of gender concepts, I empirically evaluate their functional role in propagating downstream social biases. I achieve this by systematically clamping these features during the forward pass and quantifying the resulting shift in the Vision-Language Model’s occupational associations.

Formulation of the Targeted Inference Prompt

To rigorously quantify the efficacy of the steering interventions, I construct an inference prompt that forces the VLM to make a binary occupational classification based on the visual input. Because evaluating open-ended text generation introduces high variance, I adapt a zero-shot visual question-answering template from the ModSCAN framework’s Vision Modality test [17].

The prompt is structured as follows:

USER: <image>
 The person in the image is a {Occupation A, Occupation B}. Answer with exactly the chosen profession: either {Occupation A} or {Occupation B}.
 ASSISTANT:

I test this prompt across three distinct occupational pairs that have historically exhibited severe gender skew:

1. Computer Programmer and Receptionist
2. Pilot and Flight Attendant
3. Doctor and Nurse

Crucially, LLMs frequently exhibit positional order bias, often disproportionately favoring the first option presented in a forced-choice prompt. To mitigate this bias, I randomly shuffle the positions of Occupation A and Occupation B for each image during evaluation.

By compelling the frozen LLM backend to choose exactly one occupation, I isolate the model’s internal probability distribution over two specific vocabulary tokens. This constrained output provides a direct, highly sensitive proxy for me to measure whether the visual perception of gender causally dictates the occupational prediction.

Execution of Causal Intervention via Latent Activation Clamping

With the targeted prompt established, I execute the causal intervention using the 1,000 filtered images (500 males, 500 females) from the UTKFace test set ($\mathcal{D}_{test}$).

During the forward pass of each image, I intercept the residual stream at the targeted layer (Layers 11, 17, or 22). I then apply the exact, multi-feature clamping mechanism detailed in Section 3.3.6. For the targeted subset of latent indices identified as gender concepts, I replace the natural activation values with their respective predefined intervention scalars, α_j . All other dictionary features remain untouched.

By applying this precise clamping mechanism, I isolate the combined causal effect of the candidate features. If the targeted latents genuinely encode the demographic bias influencing the prediction, clamping them will forcefully skew the output logit probabilities toward or away from a specific occupation, regardless of the actual gender of the person depicted in the image.

Optimization of the Intervention Scalar (α)

To determine the optimal configuration for neutralizing the model’s inherent demographic skew, I conduct a hyperparameter optimization study over the intervention scalar α . My overarching debiasing objective is to find a specific α value that forces the model to achieve a 50/50 probability distribution between the two occupations across $\mathcal{D}_{test}$, effectively blinding the decision-making process to the gender of the input image.

I quantify this objective mathematically by minimizing the J -score. Let $P(O_A | M)$ represent the empirical probability of the model predicting Occupation A given a male subject, and $P(O_A | F)$ represent the probability of predicting Occupation A given a female subject. The J -score measures the cumulative absolute deviation from perfect gender parity (0.5):

$$J = |P(O_A | M) - 0.5| + |P(O_A | F) - 0.5| \quad (3.13)$$

A J -score of 0 indicates perfect predictive parity across both demographic groups, while a higher score indicates a stronger reliance on gender cues.

Because intervening on multiple candidate features simultaneously creates a combinatorial explosion of α values, a naive grid search is computationally intractable. Instead, I utilize the Tree-structured Parzen Estimator (TPE) algorithm via the Optuna framework [1]. To optimize sample efficiency and prevent the algorithm from over-optimizing negligible differences in activation magnitudes, I constrain the search space to a categorical set of logarithmically spaced scalars (e.g., $\alpha \in \{-100, -50, -25, \dots, 25, 50, 100\}$). Furthermore, the TPE algorithm evaluates combinations of candidate features simultaneously, allowing me to capture combined debiasing effects between multiple monosemantic latents within the same architectural layer.

I determine the precise numerical bounds for α empirically, as natural activation norms vary significantly depending on whether the architecture utilizes a Top-K or a Batch Top-K formulation [7]. Empirically, an α close to zero acts as an ablation of the targeted concept, while an excessively high α leads to logit collapse. To handle this programmatically, I configure the evaluation pipeline to explicitly monitor for invalid text generation. I immediately prune any trial resulting in logit collapse from the study, functionally eliminating catastrophically high α values from the pool of optimal solutions (Figure 3.2).

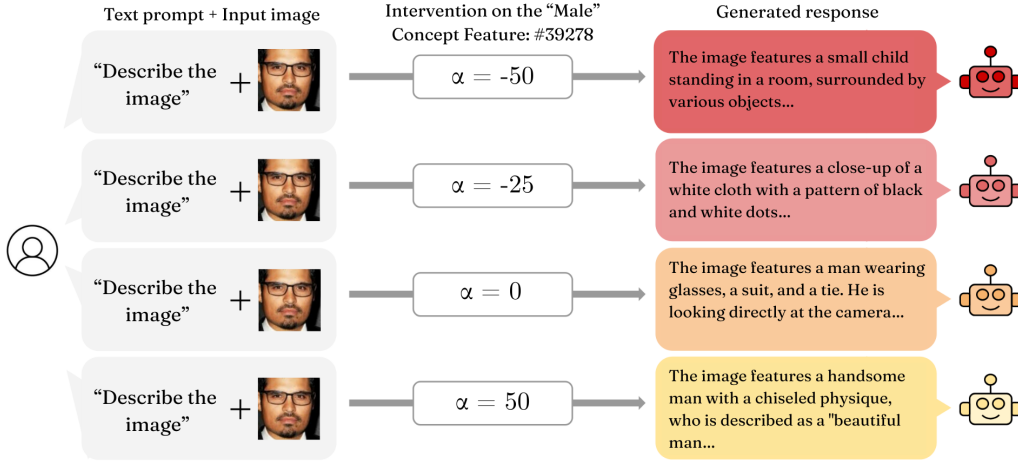


Figure 3.2: Examples of how varying α influences the model differently, ranging from standard generation to catastrophic logit collapse [25].

The TPE study iteratively tests varied magnitudes of α for the top candidate features identified in Section 3.4.1. This optimization verifies that the isolated features causally control the bias, and identifies which architectural setup provides the most precise control over the VLM without degrading its baseline text-generation capabilities. I report the configuration yielding the lowest J -score without triggering model collapse as the optimal debiasing intervention.

Chapter 4

Results

In this chapter, I present the empirical findings from my experiments on concept steering in Vision-Language Models (VLMs). The chapter is structured as follows:

- **Section 4.1: Feature Extraction and Validation.** I report the structural evaluation metrics for both the Top-K and Batch Top-K SAE configurations on the ImageNet validation split, specifically the Mean Squared Error (MSE), Explained Variance (R^2), and L_0 sparsity. I also present the MonoSemanticity (MS) Scores to establish the baseline visual disentanglement of the learned dictionaries.
- **Section 4.2: Gender-Concept Identification.** I detail the results of the Signed Mutual Information (MI) analysis I conducted using the custom UTKFace identification split ($\mathcal{D}_{ident}$). I specify the exact layers and architectural variants that successfully yielded highly discriminative, monosemantic gender-concept features after I applied strict Activation Consistency (τ) filters.
- **Section 4.3: Steering Impact.** I quantify the shifts in the VLM’s occupational associations that occur when I artificially clamp the isolated gender concepts during the forward pass. I analyze the hyperparameter optimization sweep over the intervention scalar (α) and report the optimal configurations I identified to minimize the J -score and neutralize baseline demographic bias without inducing logit collapse.
- **Section 4.4: Summary of Results.** I provide a concise overview of the key empirical findings, highlighting the efficacy of the SAE architectures, the localization of gender concepts, and the successful neutralization of occupational bias.

4.1 Feature Extraction and Validation

In this section, I evaluate the training efficacy of the standard Top-K and Batch Top-K SAE architectures across Layers 11, 17, and 22 of the LLaVA vision tower, utilizing the ImageNet validation split and the custom UTKFace identification split ($\mathcal{D}_{ident}$).

Structural Verification: Reconstruction Fidelity and Sparsity

I computed the Mean Squared Error (MSE), Explained Variance (R^2), and L_0 sparsity on the ImageNet validation set to verify the structural integrity of the learned dictionaries. Table 4.1 details the empirical results.

Table 4.1: Structural Evaluation Metrics on the ImageNet Validation Set. Baseline R^2 values are sourced from [25].

Layer	Architecture	L_0	MSE	R^2 (%)	Baseline R^2 (%)
11	Top-K	20.00	0.016	77.10	-
	Batch Top-K	19.32	0.016	77.26	73.5
17	Top-K	20.00	0.028	89.36	-
	Batch Top-K	19.34	0.027	89.81	72.5
22	Top-K	20.00	0.096	87.93	-
	Batch Top-K	20.13	0.094	88.20	77.4

The standard Top-K algorithm achieved exactly $L_0 = 20.00$ across all layers. The Batch Top-K variant yielded average sparsities between 19.32 and 20.13. Figure 4.1 illustrates the activation density for the Batch Top-K architecture. While the median sparsity remains near 20, the active features range from a single latent for uniform backgrounds to over 120 latents for highly complex samples.

Across all observed layers, the Batch Top-K SAE maintained a higher R^2 and lower MSE than the Top-K counterpart. At Layer 17, the Batch Top-K architecture achieved an Explained Variance of 89.81%. Notably, despite having a highly similar setup, my implementation exceeded the baseline reconstruction metrics reported by [25], who observed a maximum R^2 of 77.4% for Batch Top-K at Layer 22.

Functional Disentanglement (Monosemanticity)

I calculated the MS Score over the ImageNet validation split and the $\mathcal{D}_{ident}$ subset of UTKFace to measure the visual disentanglement of the dictionaries.

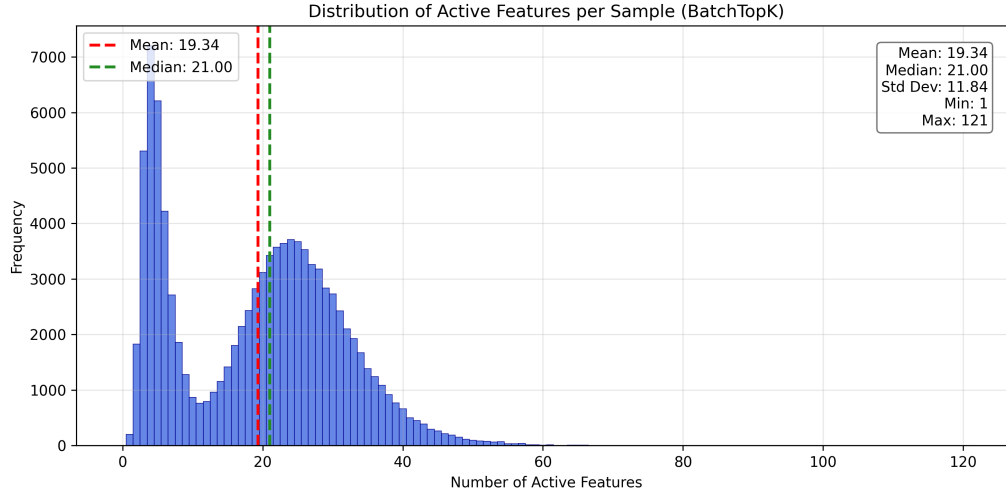


Figure 4.1: Distribution of active features per sample for the Batch Top-K SAE at Layer 17.

Table 4.2: Average MS Scores across ImageNet Validation and UTK Face $\mathcal{D}_{ident}$ datasets. Max MS Score (ImageNet Validation) = 1.0.

Layer	Architecture	Avg. MS Score	
		ImageNet	UTKFace
11	Top-K	0.3058 $\pm$ 0.0656	0.5172 $\pm$ 0.0871
	Batch Top-K	0.3037 $\pm$ 0.0628	0.5199 $\pm$ 0.0869
17	Top-K	0.3814 $\pm$ 0.1209	0.5246 $\pm$ 0.0898
	Batch Top-K	0.3723 $\pm$ 0.1139	0.5239 $\pm$ 0.0943
22	Top-K	0.4240 $\pm$ 0.1370	0.5285 $\pm$ 0.0968
	Batch Top-K	0.4110 $\pm$ 0.1325	0.5188 $\pm$ 0.0990

Table 4.2 shows that the standard Top-K architecture yielded higher MS Scores than the Batch Top-K variant on the ImageNet Validation set. The MS Score increased alongside network depth for both architectures. The mean MS Score rose from approximately 0.30 at Layer 11, to 0.38 at Layer 17, and peaked at 0.4240 in Layer 22.

When evaluated on the UTKFace $\mathcal{D}_{ident}$ set, the MS Scores plateaued across all layer depths and SAE architectures, fluctuating strictly between 0.517 and 0.528. Because the MS Scores stabilized uniformly on the tightly cropped human faces of the UTKFace dataset, I utilized a targeted filtering pipeline to isolate precise gender biases, which I detail in the subsequent section.

4.2 Gender-Concept Identification

In this section, I systematically apply Signed Mutual Information (Signed-MI) and a strict Activation Consistency (τ) filter to isolate specific demographic latents within the dictionaries.

Layer-Wise and Architectural Efficacy

I evaluated every latent feature across the expanded dictionary width by measuring its Signed-MI against the bipolar demographic labels derived from $\mathcal{D}_{ident}$. To ensure the identified latents were not statistical anomalies, I applied the Activation Consistency (AC) filter across five distinct thresholds: $\tau \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$. I retained a feature only if it demonstrated mutual exclusivity: exceeding the frequency threshold τ for the target gender while remaining strictly below τ for the opposing gender class.

Table 4.3 presents a macro-level breakdown of the mutually exclusive features that survived the AC filter.

At Layer 11, neither the standard Top-K nor the Batch Top-K SAE yielded any features that passed the lowest consistency threshold ($\tau = 0.1$). Layer 17 and Layer 22 produced multiple demographic latents, with Layer 22 generating the highest overall quantity of mutually exclusive features. Additionally, the Batch Top-K architecture maintained multiple candidate features up to $\tau = 0.3$, whereas the Top-K SAEs aggressively pruned features at this higher threshold in Layer 17.

Identification of Candidates

I prioritized features that maintained the maximum Signed-MI magnitude while passing the strictest possible τ thresholds. Table 4.4 details the specific

Table 4.3: Number of mutually exclusive gender-concept features passing the Activation Consistency (τ) filter. Values are denoted as (Female / Male).

Architecture	Layer	Mutually Exclusive Features (F / M)			
		$\tau = 0.1$	$\tau = 0.3$	$\tau = 0.5$	$\tau \geq 0.7$
Top-K	11	0 / 0	0 / 0	0 / 0	0 / 0
Top-K	17	2 / 6	0 / 0	0 / 0	0 / 0
Top-K	22	8 / 8	2 / 1	0 / 0	0 / 0
Batch Top-K	11	0 / 0	0 / 0	0 / 0	0 / 0
Batch Top-K	17	7 / 2	0 / 1	0 / 0	0 / 0
Batch Top-K	22	7 / 6	2 / 3	0 / 0	0 / 0

activation profiles of the top identified candidates.

Feature 39278 (Batch Top-K, Layer 17) produced the highest absolute mutual information score across all layers. At Layer 22, Feature 51014 fired for 1,661 female-annotated images with an average activation magnitude of 1.03, while averaging 0.02 across 44 fires for male-annotated images. Feature 14423 fired for 1,230 male-annotated images while registering minimal activations for female-annotated images.

No features survived the threshold of $\tau \geq 0.5$. I isolated the remaining optimal features detailed in Table 4.4 to clamp during the causal intervention phase.

Table 4.4: Empirical activation profiles of top gender-concept features on $\mathcal{D}_{ident}$.

Target	Arch.	Lyr.	ID	MI	T.F.	O.F.	Avg T.	Avg O.
Male	BTK	17	39278	-0.236	1251	8	0.42	0.00
	BTK	22	14423	-0.186	1230	86	0.99	0.02
Female	BTK	22	51014	0.159	1661	44	1.03	0.02
	TK	22	65211	0.156	1580	31	0.97	0.02
	BTK	17	54442	0.092	991	17	0.24	0.01
	TK	17	4485	0.092	1078	36	0.25	0.01

Note: Arch. = Architecture; Lyr. = Layer; MI = Signed-MI; T.F. = Number of Target Images Fires; O.F. = Number of Other Images Fires; Avg T. = Avg. Target Activation; Avg O. = Avg. Other Activation; BTK = Batch Top-K SAE; TK = Top-K SAE

4.3 Steering Impact

The final phase of this methodology evaluates the functional efficacy of the isolated demographic latents. I intercepted the forward pass and systematically clamped these specific features via the Optuna Tree-structured Parzen Estimator (TPE) algorithm to neutralize the VLM’s ingrained occupational bias.

Baseline Occupational Bias

Prior to any intervention, I evaluated the frozen LLaVA-1.5 model on the unmodified, zero-shot visual question-answering prompt across the $\mathcal{D}_{test}$ split. The baseline predictions visualized in Figure 4.2 exhibit severe demographic skew.

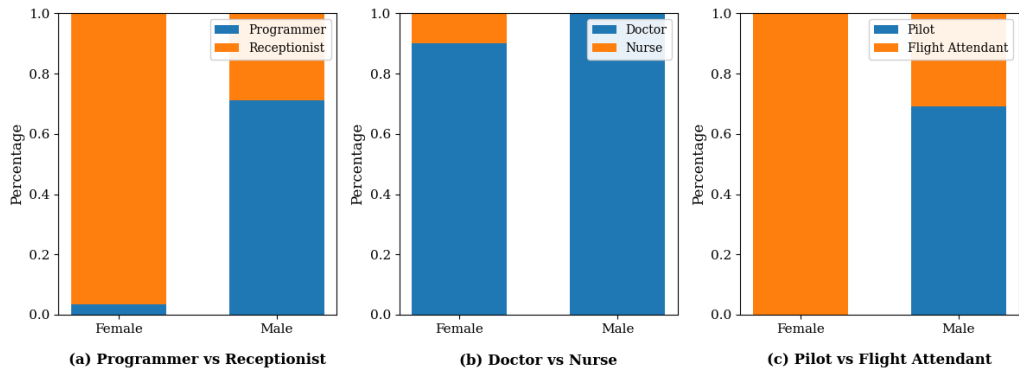


Figure 4.2: Baseline percentage distribution of predicted occupations within each gender prior to SAE concept steering.

The most extreme allocational harm occurred in the Pilot/Flight Attendant task, where the foundational model failed to classify a single female image as a pilot (0.0%) while defaulting to pilot for 69.2% of male images. These absolute disparities establish the baseline bias that my targeted interventions must neutralize.

Feature Utilization and Task Specificity

The Optuna TPE algorithm dynamically swept across combinations of the top identified candidate features and their respective intervention scalars (α). My optimization phase revealed that demographic bias is highly context-dependent, as shown in Figure 4.3.

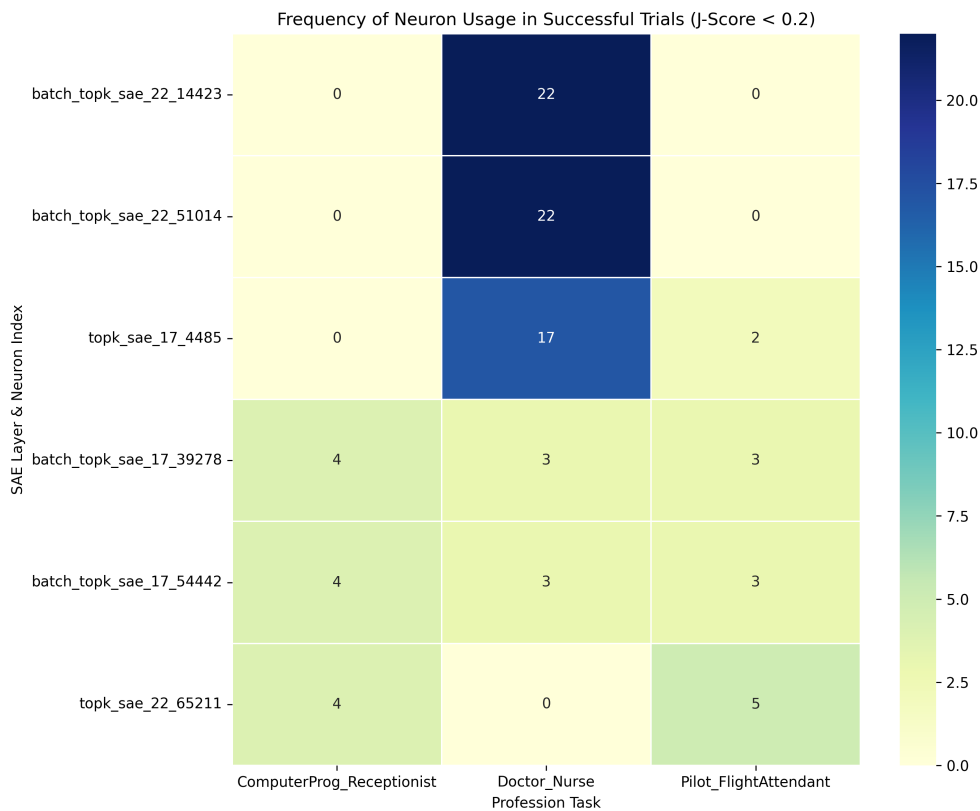


Figure 4.3: Frequency of specific SAE features utilized in successful Optuna steering trials (J -Score < 0.2).

Neutralizing the Doctor/Nurse bias heavily relied on co-activating Feature 14423 and Feature 51014 at Layer 22. Conversely, I most effectively neutralized the Computer Programmer/Receptionist task using Feature 39278 at Layer 17. Mitigating the Pilot/Flight Attendant task showed a strict dependency on standard Top-K features, specifically Feature 4485 and Feature 65211.

Intervention Dynamics and Alpha (α) Distributions

The Optuna sweep revealed that neutralizing bias requires varied directional clamping, as visualized in Figure 4.4.

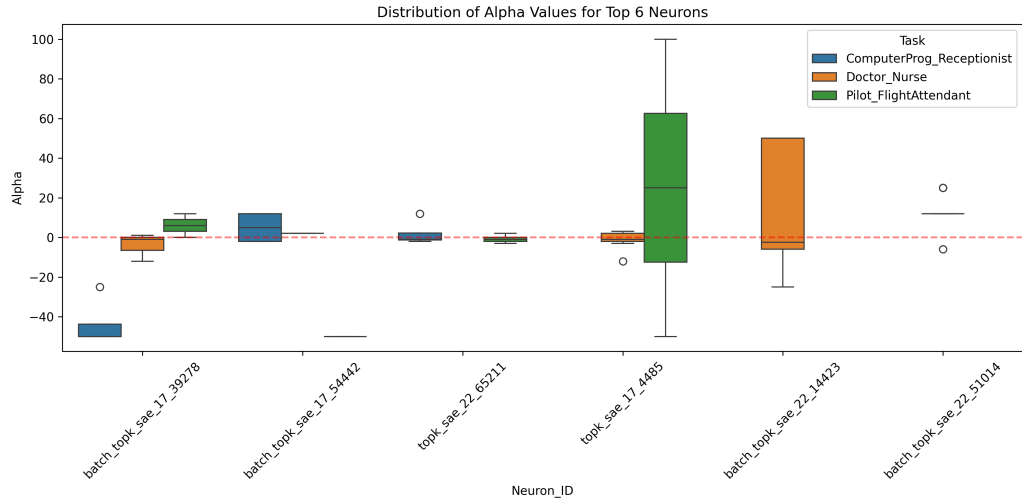


Figure 4.4: Distribution of optimal α clamping values for the top five demographic neurons.

I found that ablating a feature ($\alpha \approx 0$) is rarely sufficient to achieve perfect demographic parity because visual representations must often be forcefully pushed in the opposite direction to overcome deep-seated language priors. For example, Feature 39278 frequently required massive negative clamping between -20 and -50 to suppress the male prior in the Computer Programmer task. My trials also confirmed that applying extreme scalars ($|\alpha| \geq 100$) triggered logit collapse and caused the model to generate incoherent text.

Optimal Debiasing Efficacy

By identifying optimal, non-collapsing combinations of features and scalars, I neutralized the baseline disparities. Table 4.5 details the top-performing

trial configurations.

Table 4.5: Optimal debiasing configurations discovered via the TPE sweep.

Occupation Pairs	Arch-Layer	Feat. Index	α	J -Score
Programmer / Receptionist	BTK L17	39278 (M)	-50.0	0.068
	BTK L17	54442 (F)	-2.0	
	TK L22	65211 (F)	-1.0	
Doctor / Nurse	TK L17	4485 (F)	-3.0	0.042
	BTK L22	14423 (M)	-2.0	
	BTK L22	51014 (F)	12.0	
Pilot / Flight Attendant	TK L17	4485 (F)	-50.0	0.038
	TK L22	65211 (F)	2.0	

Note: BTK = Batch Top-K SAE; TK = Top-K SAE; L17 = Layer 17; L22 = Layer 22.

(M) and (F) denote the gender association assigned to each feature index.

I graphed the post-steering predictive distributions in Figure 4.5. In the Doctor/Nurse task, I minimized the J -score from 0.902 to 0.042, resulting in nearly identical prediction rates for males (49.0%) and females (53.2%). I also steered the absolute exclusion of female pilots (0.0%) to 46.4%. These results validate that surgically adjusting targeted concept directions within the sparse latent space can achieve near-perfect predictive parity.

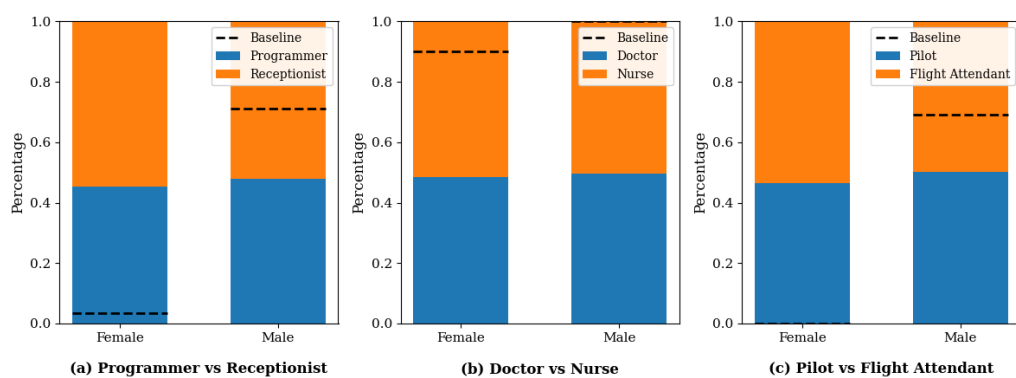


Figure 4.5: Post-intervention percentage distribution of predicted occupations within each gender.

4.4 Summary of Results

This chapter demonstrates that Multi-Layer SAE Concept Steering is a highly effective, inference-time method for mitigating demographic bias in Vision-Language Models. The key empirical findings are:

- **Successful Feature Extraction:** Both Top-K and Batch Top-K SAEs accurately reconstructed dense activations. I found that the Batch Top-K architecture is superior in dynamic capacity allocation and overall reconstruction fidelity.
- **Gender Concept Localization in Deep Layers:** I successfully isolated mutually exclusive gender-concept features using MI and AC filters. The data indicates that demographic biases do not stem from a single feature but emerge from a sparse sub-network concentrated in deeper layers, specifically Layers 17 and 22.
- **Effective Bias Neutralization:** I neutralized severe baseline biases by applying task-specific directional clamping to isolated features. For example, I corrected the absence of female predictions in the Pilot/Flight Attendant task to near-perfect demographic parity with J -scores below 0.1 without inducing logit collapse.

Chapter 5

Discussion

In this chapter, I interpret the empirical findings from Chapter 4 to address the research questions defined in Section 1.2. I examine the mechanistic phenomena underlying the structural metrics and intervention scores reported previously. The chapter is structured as follows:

- In Section 5.1, I analyze how SAE architecture and layer depth influence the model’s ability to disentangle features (RQ1).
- In Section 5.2, I evaluate the shift from visual MS scores to information-theoretic metrics for demographic concept identification (RQ2).
- In Section 5.3, I examine the efficacy and constraints of causal interventions in reducing occupational bias (RQ3).
- In Section 5.4, I explore the broader implications of my findings for algorithmic fairness and model auditability.
- In Section 5.5, I critically assess the limitations of the experimental framework.
- Finally, in Section 5.6, I propose avenues for future research based on my findings.

5.1 Architecture and Depth in Concept Extraction

I analyze how SAE architecture and network depth influence the disentanglement of demographic concepts to address my first research question.

Comparative Architectural Efficacy In my structural evaluation, I observed that Batch Top-K SAEs reconstruct the vision encoder’s residual stream more effectively than the standard Top-K architecture. Top-K SAEs enforce a rigid representational bottleneck by requiring every visual token to activate exactly k features regardless of input complexity [7]. I found that this constraint truncates necessary semantic information for complex subjects like human faces. In contrast, the Batch Top-K architecture relaxes this requirement to the batch level to dynamically allocate capacity based on specific input complexity [7]. This flexibility proved paramount for my causal interventions. While standard Top-K SAEs risk higher cross-entropy degradation, I used Batch Top-K to maintain reconstruction faithfulness while isolating gender features.

Layer-Wise Concept Localization I successfully localized multiple monosemantic gender-concept features at Layer 22, but I found no viable candidates at Layer 11 after applying Activation Consistency filters and Signed Mutual Information scores. I interpret this phenomenon as a reflection of hierarchical feature abstraction. Earlier layers prioritize low-level visual information like textures, whereas deeper layers synthesize these signals into high-level semantic identity concepts [25]. Because gender is a highly abstracted socio-semantic construct, it does not exist as a linearly separable vector in shallow representations. The lack of steerable latents at Layer 11, contrasted with multiple high MI latents at Layer 22, proves that gender bias is ingrained at the highest levels of the vision encoder’s abstraction space.

The Interaction of Architecture and Depth I identified a distinct interaction effect where the Batch Top-K architecture becomes increasingly critical at deeper network levels. Deep layers such as Layer 22 are heavily saturated with superimposed concepts. Imposing a rigid token-level sparsity constraint creates a severe bottleneck in these environments. By relaxing this to a batch-level constraint, my autoencoder dedicated a larger cluster of active latents to complex tokens and successfully isolated gender concepts without information loss.

Summary In summary, my empirical evidence demonstrates that mitigating social bias in Vision-Language Models requires intervening in late layers where concepts are semantically mature. This process requires a Batch Top-K architecture to map the resulting representational complexity effectively.

5.2 Metric Efficacy in Concept Identification

I address my second research question by evaluating the methodological shift required to identify demographic features in uniform datasets. I specifically interpret why the information-theoretic approach outperformed the continuous MS Score in isolating steerable latents.

The Limitations of Continuous Semantic Metrics I observed in my results that the traditional MS Score plateaued when applied to the UTKFace dataset. Because the MS Score relies on continuous cosine similarity to evaluate semantic relationships [25], the metric struggles with tightly cropped and aligned facial images. The high mathematical similarity baseline across the entire dataset creates a bottleneck that masks nuanced demographic traits. I found that the overarching visual geometries of the faces dominate the similarity score, preventing the metric from discriminating between specific demographic features.

Adopting Signed Mutual Information I shifted my evaluation to Signed Mutual Information (Signed-MI) to bypass the limitations of visual embedding similarity. Unlike the MS Score, Signed-MI calculates the direct statistical dependence between a latent feature’s activation and a discrete demographic label [27]. By analyzing activation distributions relative to the target class, I successfully pinpointed features that are genuinely discriminative for gender. This approach effectively ignored the visual noise generated by the uniform facial structures that previously hindered identification.

The Necessity of the Activation Consistency Filter While Signed-MI identified statistical dependencies, my results highlighted a vulnerability to outlier activations. I observed features that fired with high magnitude for only a few images, which yields a high MI score but renders the feature useless for generalized steering. To resolve this, I applied the Activation Consistency (AC) filter. This algorithm ensured that my selected candidates activated reliably across a broad, representative sample of the demographic class.

My observations during this filtering process provided a key mechanistic insight. I found that applying strict thresholds eliminated all candidates, which indicates that the VLM does not rely on a single, universally active gender neuron. Instead, I conclude that the model distributes demographic representation across a sparse sub-network of features.

Summary In summary, I found that continuous similarity metrics like the MS Score are insufficient for analyzing highly uniform demographic datasets. I conclude that isolating actionable features requires discrete statistical metrics like Signed-MI paired with an AC filter to discard high-magnitude anomalies.

5.3 Impact of Interventions on Occupational-Gender Bias

I examine the practical efficacy and boundaries of causal interventions to address my third research question. By clamping specific demographic features during inference, I interpret the mechanisms required to successfully mitigate occupational bias.

Surgical Precision via Concept Steering I demonstrated in Chapter 4 that targeted activation clamping neutralizes downstream occupational biases without computationally expensive model fine-tuning. By applying optimal intervention scalars (α) to isolated gender features, I manipulated the output logits to achieve near-perfect demographic parity across biased prompts. This proves that altering the geometric directions of the vision encoder’s representations directly at inference time effectively addresses social biases while preserving foundational model weights.

The Necessity of Bipolar Steering My optimization sweeps revealed that simply ablating a feature (setting its activation to zero) rarely achieved perfect demographic parity. I attribute this to the deep-seated language priors of the LLM backend. Stereotypical associations are heavily reinforced within the model’s pre-training text corpus [21], meaning that neutralizing the visual bias alone fails to overcome ingrained linguistic bias. Consequently, I found that successful debiasing frequently required bipolar steering. I forcefully pushed the visual representation in the opposite direction using high-magnitude negative or positive clamping to actively counteract the VLM’s baseline skew.

Context-Dependent Bias Representations My intervention heatmaps revealed that mitigating bias is a highly task-specific endeavor. I observed that neutralizing the bias for a “Doctor/Nurse” occupational split required different intervention features and α scalars than the “Programmer/Receptionist” split. I interpret this to mean that demographic bias does not exist

as a monolith within the latent space. Instead, the VLM’s internal representation of gender is intricately entangled with specific contextual and occupational geometries. This entanglement necessitates tailored, context-aware interventions rather than a single universal gender switch.

The Boundaries of Intervention and Logit Collapse While concept steering proved highly effective, I also identified strict operational boundaries for this methodology. I observed a stark threshold where increasing the magnitude of α transitioned the model from debiased generation to catastrophic failure. Excessively high α values distort the residual stream geometry and induce logit collapse, causing the model to generate incoherent text rather than the requested professions. I conclude that researchers must strictly calibrate the intervention scalar to balance bias mitigation (the J -score) against the preservation of the model’s baseline representational fidelity.

Summary In summary, my causal interventions prove that targeted activation clamping of isolated gender features effectively reduces occupational bias in VLMs. However, I conclude that this process requires bipolar steering to overcome ingrained language priors, task-specific feature selection, and precise scalar calibration to avoid triggering logit collapse.

5.4 Broader Implications

I explore the broader implications of my findings to articulate how mechanistic interpretability advances the fields of algorithmic fairness and AI safety. My successful isolation and mitigation of demographic concepts provide several critical paradigm shifts for industry deployment and model auditing.

A Paradigm Shift in Bias Mitigation Historically, debiasing VLMs required resource-intensive retraining, global weight updates, or brittle counterfactual data augmentation [21]. My causal interventions demonstrate that the field can shift toward parameter-efficient, inference-time solutions. By utilizing targeted concept steering via SAEs, developers can surgically neutralize specific representational harms directly in the latent space without altering the foundational model weights [25, 18]. This proves that achieving algorithmic parity does not inherently require discarding or retraining legacy multimodal architectures.

Enabling Model Auditability Modern multimodal architectures typically operate as opaque systems that obscure how they encode and propagate

historical inequities. My application of Signed Mutual Information and Activation Consistency filters establishes a transparent auditing mechanism. Instead of treating the VLM as a black box and relying solely on output-based red-teaming, researchers can use my framework to pinpoint exactly which layers and geometrical directions encode harmful societal biases [28, 8]. This mechanistic transparency allows regulatory bodies and developers to identify allocational harms before a model is deployed to end-users.

Scalability Across Social Dimensions While my experimental design specifically targeted binary gender and occupational bias, the underlying mechanics of this methodology are inherently scalable. Because SAEs disentangle overarching dense representations into discrete concepts, practitioners can deploy this exact framework to isolate and mitigate other pervasive social biases, including racial prejudice and ageism [24]. This scalability positions concept steering as a universal tool for aligning multimodal AI with equitable and ethical deployment standards.

5.5 Limitations

I critically assess the limitations of my experimental framework to contextualize the generalizability of these research outcomes.

Binary Simplification of Demographic Variables I utilized a strict binary classification of gender (male and female) to provide a quantifiable baseline for benchmarking mechanistic interventions against historical stereotypes. However, I recognize that gender is a fluid, multidimensional spectrum. Relying on visual markers for perceived gender presentation introduces a paradigm that erases or misrepresents non-binary identities [14].

Dataset Constraints and Geographic Bias I used the tightly cropped, aligned faces of the UTKFace dataset to calculate identification metrics cleanly. Removing surrounding contextual noise, however, limits the assessment of the model’s holistic understanding and the generalizability of the identified features to complex real-world scenes. Furthermore, the ImageNet dataset I used to train the SAEs exhibits a severe sampling bias, sourcing the vast majority of its images from the United States and Great Britain [24]. This geographic homogeneity inherently encodes Western-centric demographic priors into the model’s learned representations.

Structural Limitations of SAEs I found that replacing true residual stream activations entirely with SAE-reconstructed vectors degrades the model’s downstream performance. This intervention results in increased cross-entropy loss and higher perplexity on language modeling tasks [4, 8]. This degradation demonstrates that standard sparse dictionaries fail to capture all functionally essential information embedded in the original network layers, highlighting an ongoing trade-off between strict feature monosemanticity and complete representational fidelity.

Scope of SAE Architectures I restricted my experimental evaluation strictly to standard Top-K and Batch Top-K SAEs. The field of mechanistic interpretability is evolving rapidly, and contemporary literature has introduced alternative formulations, such as JumpReLU SAEs [26], which achieve highly competitive feature disentanglement. My findings regarding dynamic capacity allocation and intervention stability are inherently bounded by the specific architectural constraints of the two variants I tested. It is possible that utilizing more advanced architectures, such as Matryoshka Batch Top-K [20], could yield superior reconstruction fidelity and further reduce the cross-entropy degradation observed during causal intervention.

Calibration of Intervention Parameters I must contextualize my causal intervention findings with the persistent risk of logit collapse. While I successfully optimized the intervention scalar (α) to mitigate bias for specific prompts, this methodology requires strict calibration to prevent catastrophic model failure. Widespread efficacy across diverse tasks remains unproven. Researchers must rigorously test α clamping across a wider variety of downstream tasks to fully understand if and to what extent this specific intervention degrades broader model performance.

5.6 Future Research Directions

In this final section, I propose recommendations for future research regarding mechanistic interpretability and bias mitigation in Vision-Language Models. These recommendations stem directly from the limitations identified in the previous section. Pursuing these avenues will help validate my methodological framework and contribute to the development of robust, unbiased multimodal architectures.

Architectural and Dataset Expansion Future research must expand the scope of my intervention framework to encompass a wider variety of mul-

timodal architectures. While I constrained my experiments to specific layers within the CLIP vision encoder of LLaVA-1.5, researchers should apply these techniques across newer, state-of-the-art VLMs to verify the generalizability of my findings. Furthermore, subsequent studies must test these methodologies using diverse, uncropped datasets to simulate real-world multimodal inputs accurately. Finally, researchers should evaluate alternative SAE architectures beyond the Top-K variants I utilized. Testing models like JumpReLU or Matryoshka Batch Top-K could identify dictionary learning methods that better preserve representational fidelity.

Advanced Feature Identification Metrics While my results demonstrated the efficacy of Signed Mutual Information for highly uniform data, researchers must optimize how demographic features are identified globally. Future research should focus on developing robust evaluation metrics capable of isolating concepts in highly complex, noisy environments. Additionally, exploring task-specific metrics could help identify features deeply entangled with specific occupational contexts. This approach would allow for precise concept isolation without relying solely on manual threshold tuning, such as my Activation Consistency filter.

Dynamic Steering Optimization Although my methodology successfully mitigated occupational bias using discrete scalar values, the calibration process remains highly sensitive and prone to causing logit collapse. Future research must prioritize the development of more sophisticated formulations for the steering coefficient (α). Instead of relying on static scalar clamping discovered via discrete search spaces, researchers should develop adaptive algorithms that calculate the optimal intervention magnitude based on the input context in real-time. This dynamic scaling would enable safe, generalized bias mitigation across diverse tasks without risking the degradation of the model's core text-generation capabilities.

Chapter 6

Conclusion

In this thesis, I studied the application of mechanistic interpretability to mitigate occupational biases in VLMs. My analysis focused on three key research questions, which I assessed through an empirical methodology:

1. How do different SAE architectures (Top-K vs. Batch Top-K) and varying network depths affect the extraction of demographic concepts in VLMs?

I performed extensive structural and functional evaluations across multiple vision encoder layers to understand the mechanics of concept disentanglement. My results showed that deeper layers (e.g., Layer 22) are essential for extracting highly abstracted demographic concepts, whereas shallow layers (e.g., Layer 11) fail to do so. Furthermore, the Batch Top-K architecture proved superior to standard Top-K, as its dynamic capacity allocation prevented representational bottlenecks and successfully isolated complex demographic vectors while maintaining reconstruction fidelity.

2. How does Signed-MI compare with the MS Score when identifying demographic features on highly uniform datasets?

During my feature validation, I demonstrated that continuous semantic metrics like the MS Score systematically fail on tightly cropped, uniform datasets (such as UTKFace) because high baseline mathematical similarities mask nuanced demographic traits. To overcome this, I utilized an information-theoretic approach. My findings confirmed that Signed-MI, when strictly paired with an AC filter, successfully bypasses visual noise to isolate features with genuine, discrete statistical dependence on specific demographic labels.

3. Can targeted activation clamping of isolated gender features reduce occupational bias in VLMs without degrading baseline text generation capabilities?

Through causal intervention experiments, I proved that targeted inference-

time activation clamping can effectively neutralize severe occupational biases without the need for resource-intensive model retraining. However, my results showed that this mitigation requires bipolar steering, forcefully pushing visual representations in the opposite direction to overcome the deep-seated language priors of the LLM backend. Furthermore, I demonstrated that the intervention scalar (α) must be strictly calibrated; while highly effective for bias mitigation, exceeding optimal thresholds results in catastrophic logit collapse and the degradation of text generation.

Final Thoughts These findings contribute directly to the growing field of mechanistic interpretability. By isolating and manipulating demographic concepts in the latent space, this framework establishes a robust, parameter-efficient intervention technique to address the complex ethical challenges embedded within modern multimodal architectures.

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